Executive Summary — Surrogate Model Validation Report

Use case: UCFATIGUE_1 | 01 September 2026 | 2 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: GradientBoosting

Avg R^2: 0.9643 — Excellent

Q90 Accuracy — Partially met

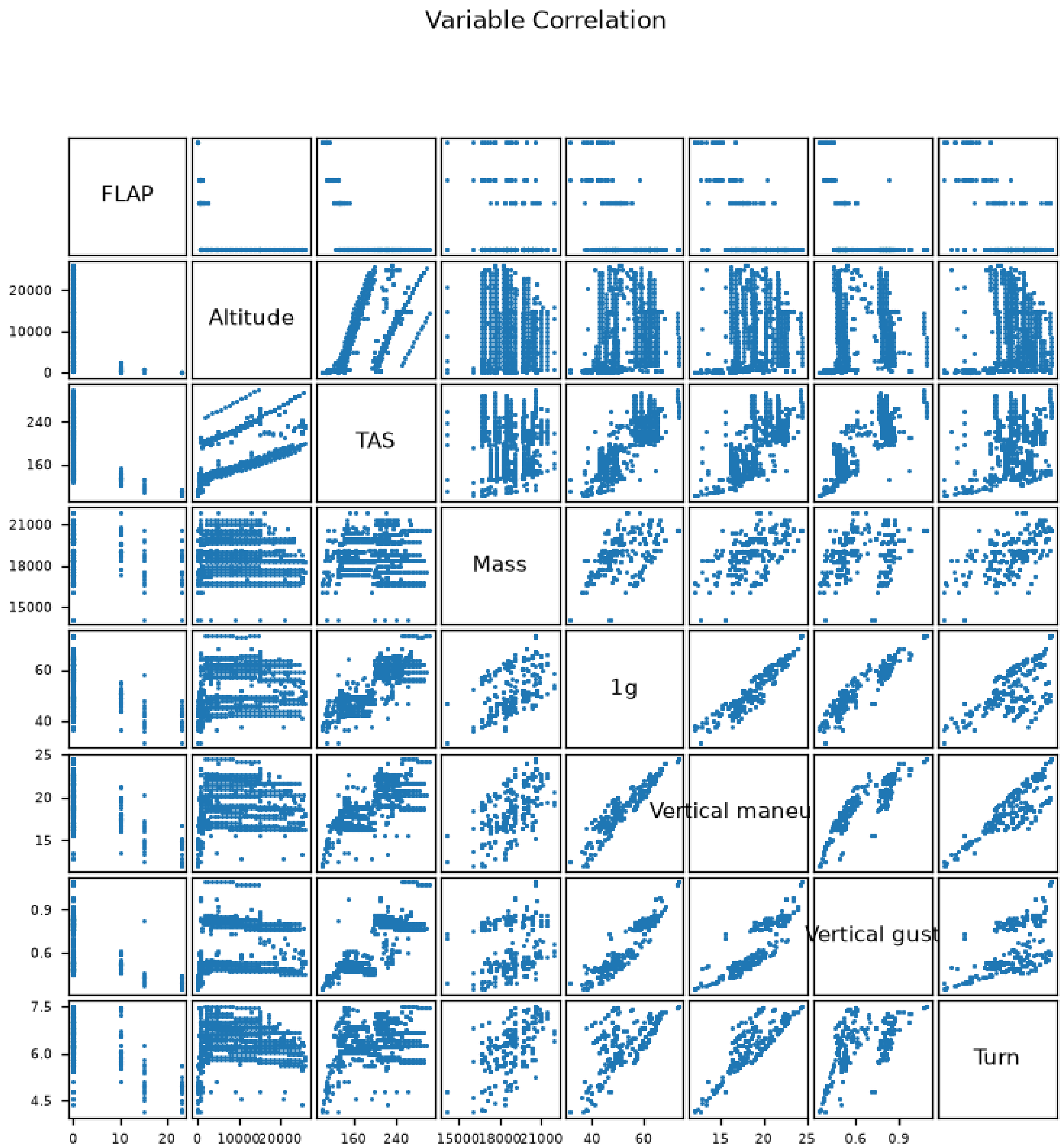
Pass: 1g, Vertical maneuver, Vertical gust, Turn, n0

Fail: Frontal gust, Giro

Project Overview

Use case	UCFATIGUE_1
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Outputs	7: 1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro
Train / Val / Test	80% / 10% / 10%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output



Model Selection

Model	Avg R^2	Q90 pass (7)	KS pass (7)
MLP	0.5237	5/7	7/7
GradientBoosting ⋆	0.9643	5/7	2/7

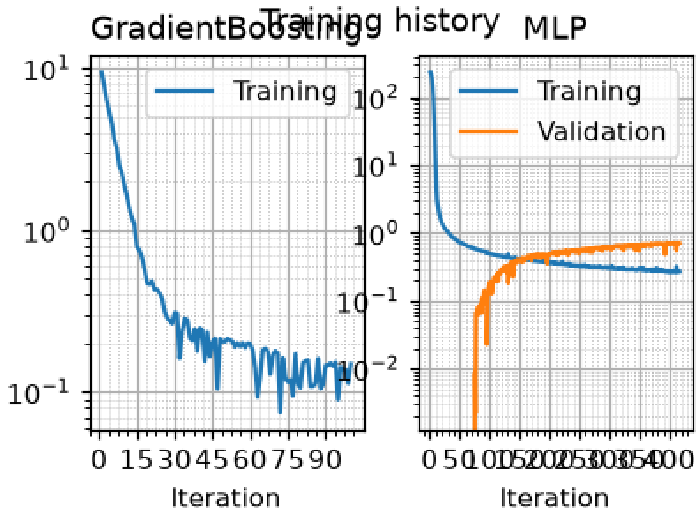
⋆ Recommended model. Q90 target: $< 10\%$. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

Output	MLP			GradientBoosting ⋆		
	R^2	Q90	Gap	R^2	Q90	Gap
1g	0.9160	0.0473	-0.0527	0.9797	0.0257	-0.0743
Vertical maneuver	0.9116	0.0554	-0.0446	0.9723	0.0318	-0.0682
Vertical gust	0.9377	0.0918	-0.0082	0.9914	0.0262	-0.0738
Turn	0.7544	0.0783	-0.0217	0.9286	0.0351	-0.0649
Frontal gust	0.9794	11.3300	+11.2300	0.9964	0.2795	+0.1795
n0	-0.7292	0.0723	-0.0277	0.9317	0.0149	-0.0851
Giro	-0.1042	0.9005	+0.8005	0.9499	0.2215	+0.1215

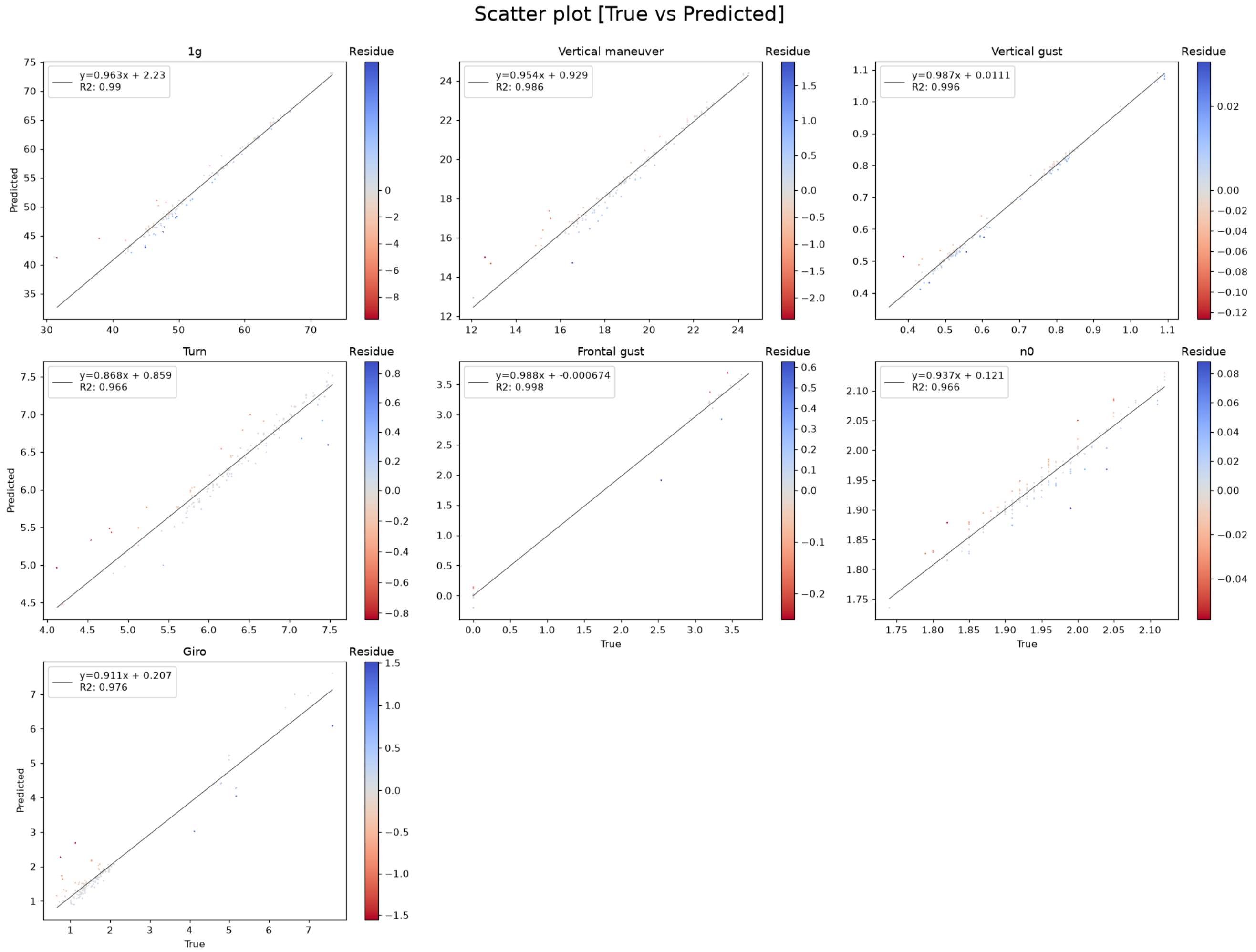
$R^2 \geq 0.95$ $R^2 \in [0.80, 0.95)$ $R^2 < 0.80$ or Q90 fail pass Gap=Q90−target ($<0 \Rightarrow$ margin)

Training History



Training loss per iteration, log scale. A validation panel is shown for models trained with early stopping. Gradient boosting reports per-stage deviance averaged over its per-output estimators.

Predicted vs True — GradientBoosting



Data Split Quality

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.931	
Phacking test proportion	0.006	
Isolated test proportion	0.063	
Chi^2 p-value (≥ 0.05)	0.7969	✓

Overfitting Check (KS Test) — GradientBoosting

Output	KS p-value
1g	0.004
Vertical maneuver	0.161
Vertical gust	0.042
Turn	0.044
Frontal gust	0.974
n0	0.032
Giro	0.017

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Deep Analysis

UCFATIGUE_1

Winner Model — Full Validation

Deep Analysis — GradientBoosting

Mirrors `validation_template.ipynb` — all computations from validation CSVs

Data Overview

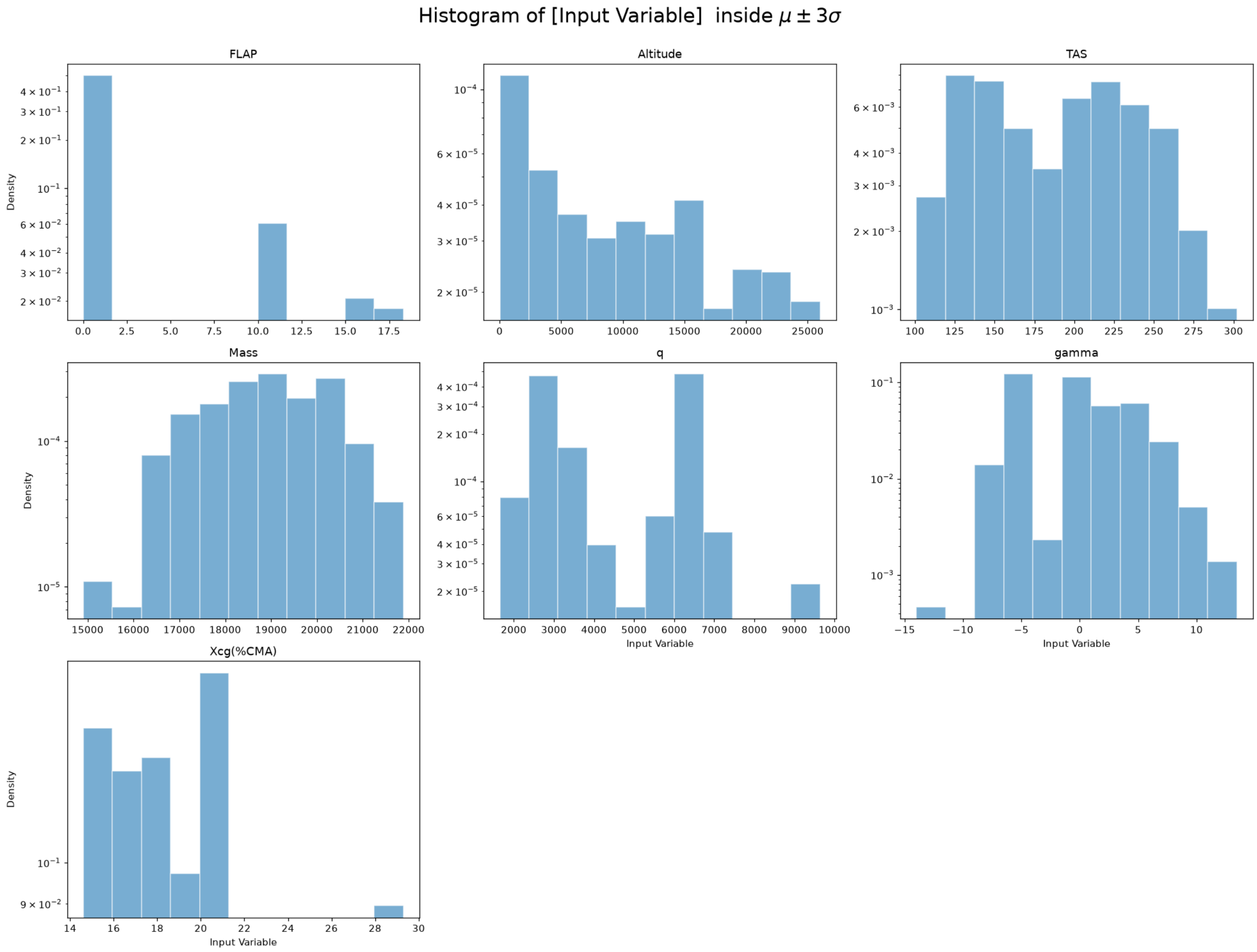
Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

Input Statistics

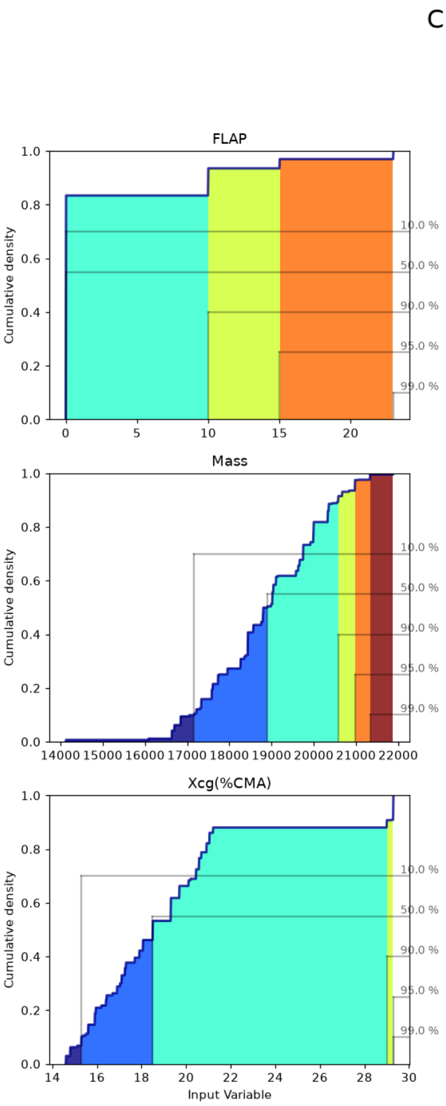
	count	mean	std	min	P25	P50	P75	max
FLAP	869	2.219	5.37	0	0	0	0	23
Altitude	869	9295	7702	18	2000	8000	1.5e+04	2.6e+04
TAS	869	190.9	48.87	101	144	197.6	230	302
Mass	869	1.887e+04	1325	1.412e+04	1.776e+04	1.89e+04	1.999e+04	2.189e+04
q	869	4510	1845	1652	2881	3580	6375	9638
gamma	869	-0.2667	4.573	-39.45	-4.704	0	2.944	18.22
Xcg(%CMA)	869	19.41	4.095	14.6	16.43	18.5	20.58	29.3

Input variables — Train_set.describe()

Input Histograms



Input Cumulative Distributions

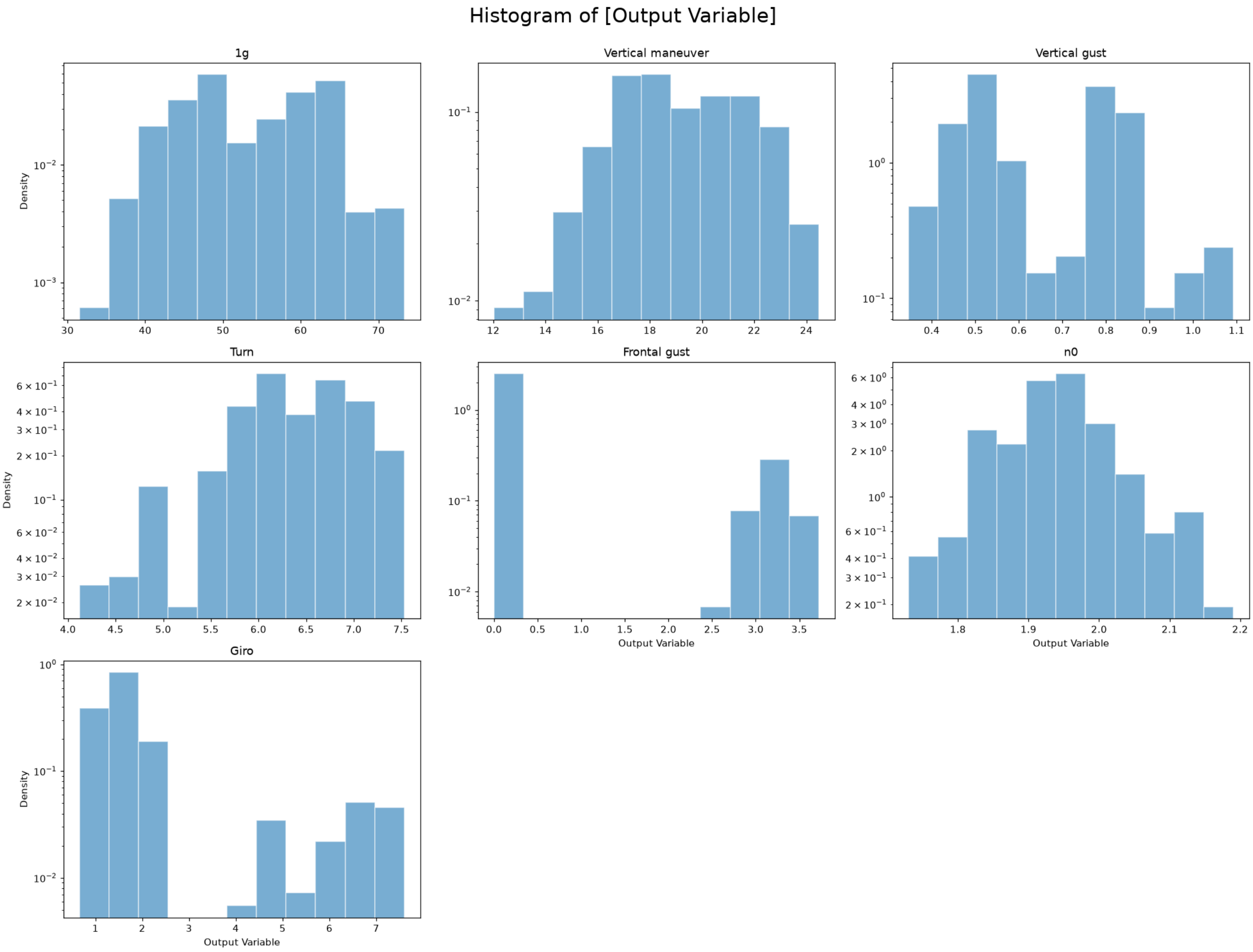


Output Statistics

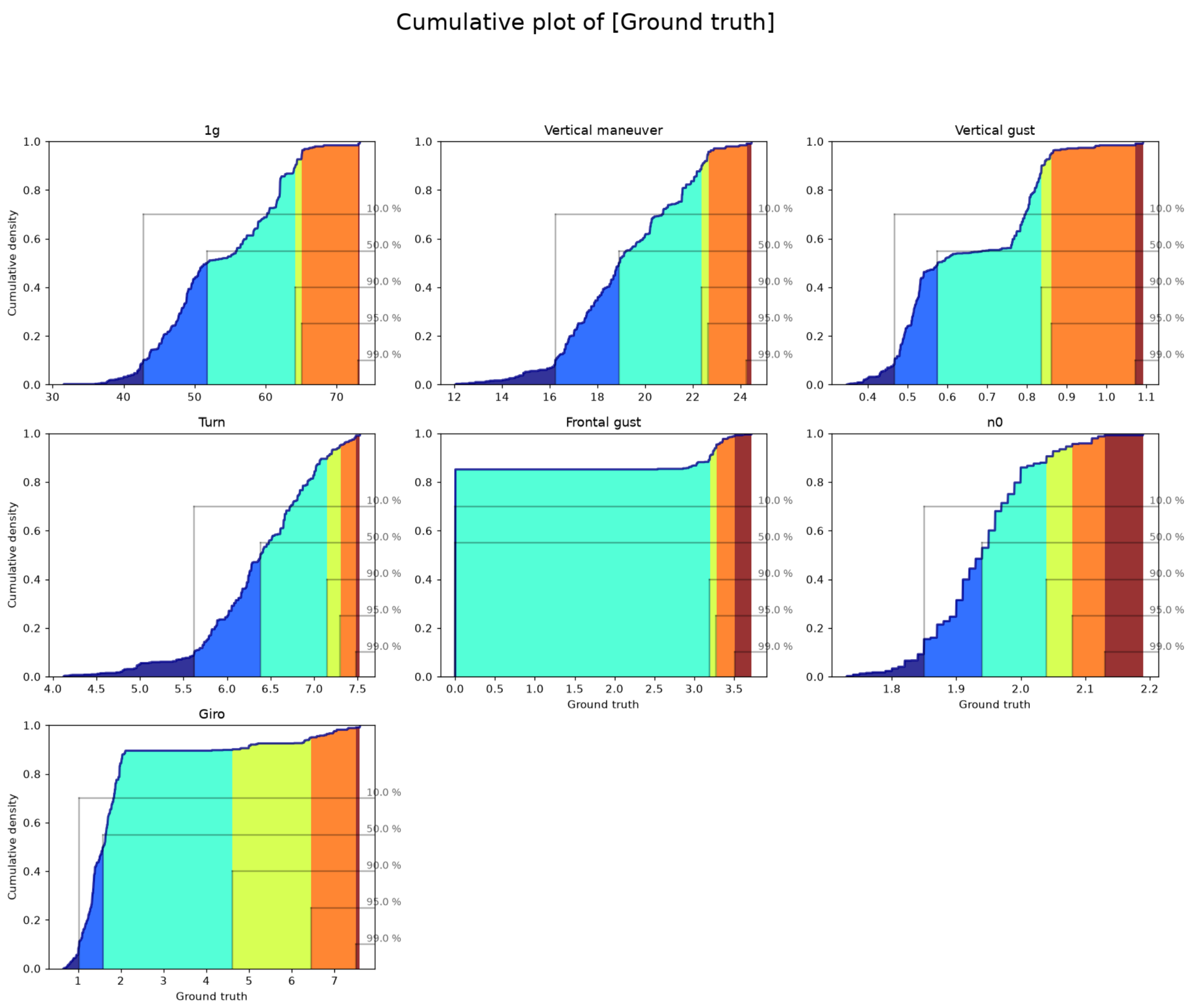
	count	mean	std	min	P25	P50	P75	max
1g	869	53.62	8.296	31.56	47.36	51.78	61.56	73.31
Vertical maneuver	869	19.14	2.454	12.03	17.18	18.89	21.39	24.47
Vertical gust	869	0.6494	0.1681	0.3466	0.5097	0.575	0.806	1.092
Turn	869	6.362	0.6415	4.122	6.01	6.382	6.83	7.53
Frontal gust	869	0.4806	1.154	0	0	0	0	3.722
n0	869	1.939	0.07768	1.73	1.9	1.94	1.98	2.19
Giro	869	1.996	1.533	0.662	1.299	1.587	1.874	7.599

Output variables — Train_set.describe()

Output Histograms



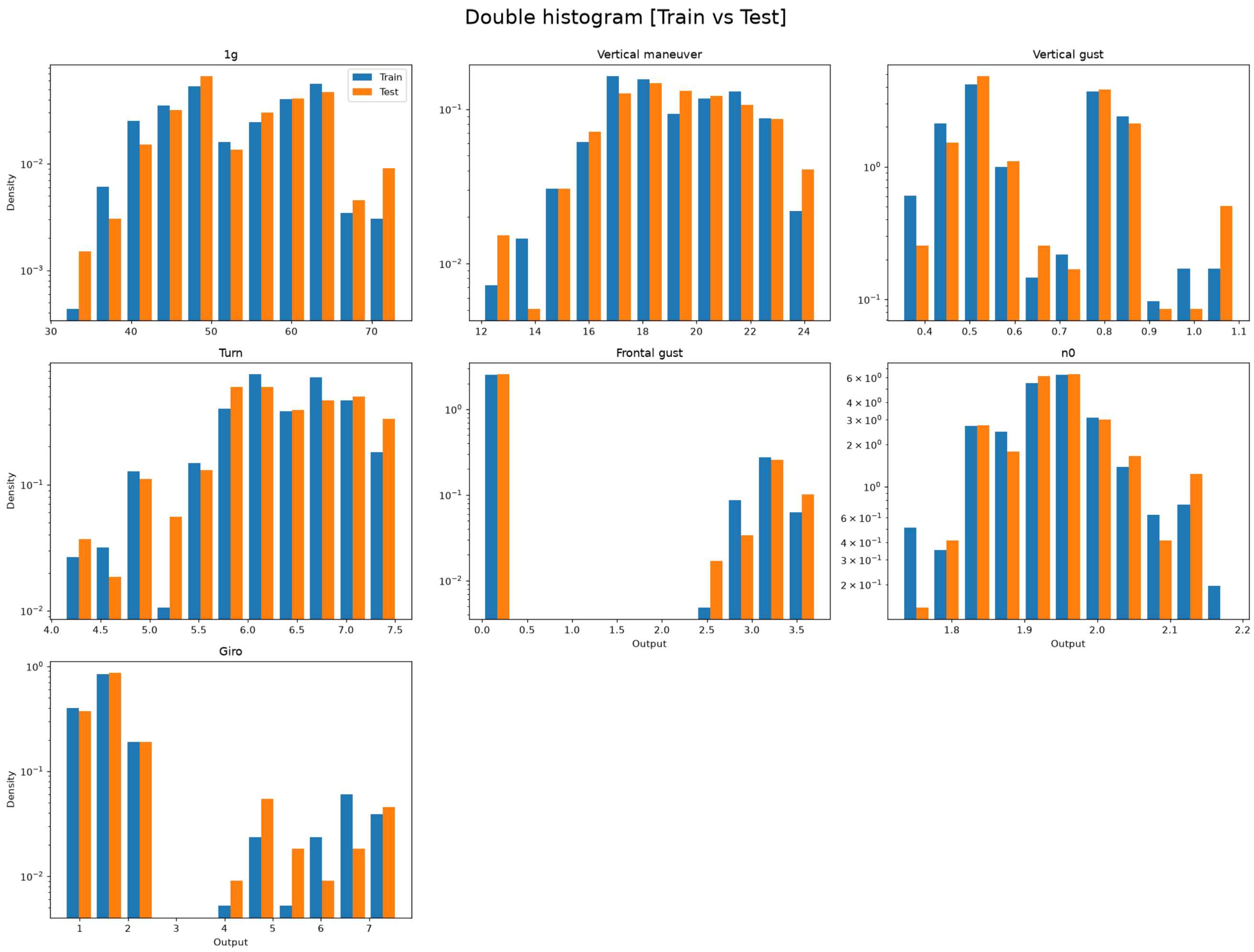
Output Cumulative Distributions


Cumulative plot of [Ground truth]
1g
Vertical maneuver
Vertical gust
Turn
Frontal gust
n0
Giro
Cumulative density
Ground truth
10.0 %
50.0 %
90.0 %
95.0 %
99.0 %

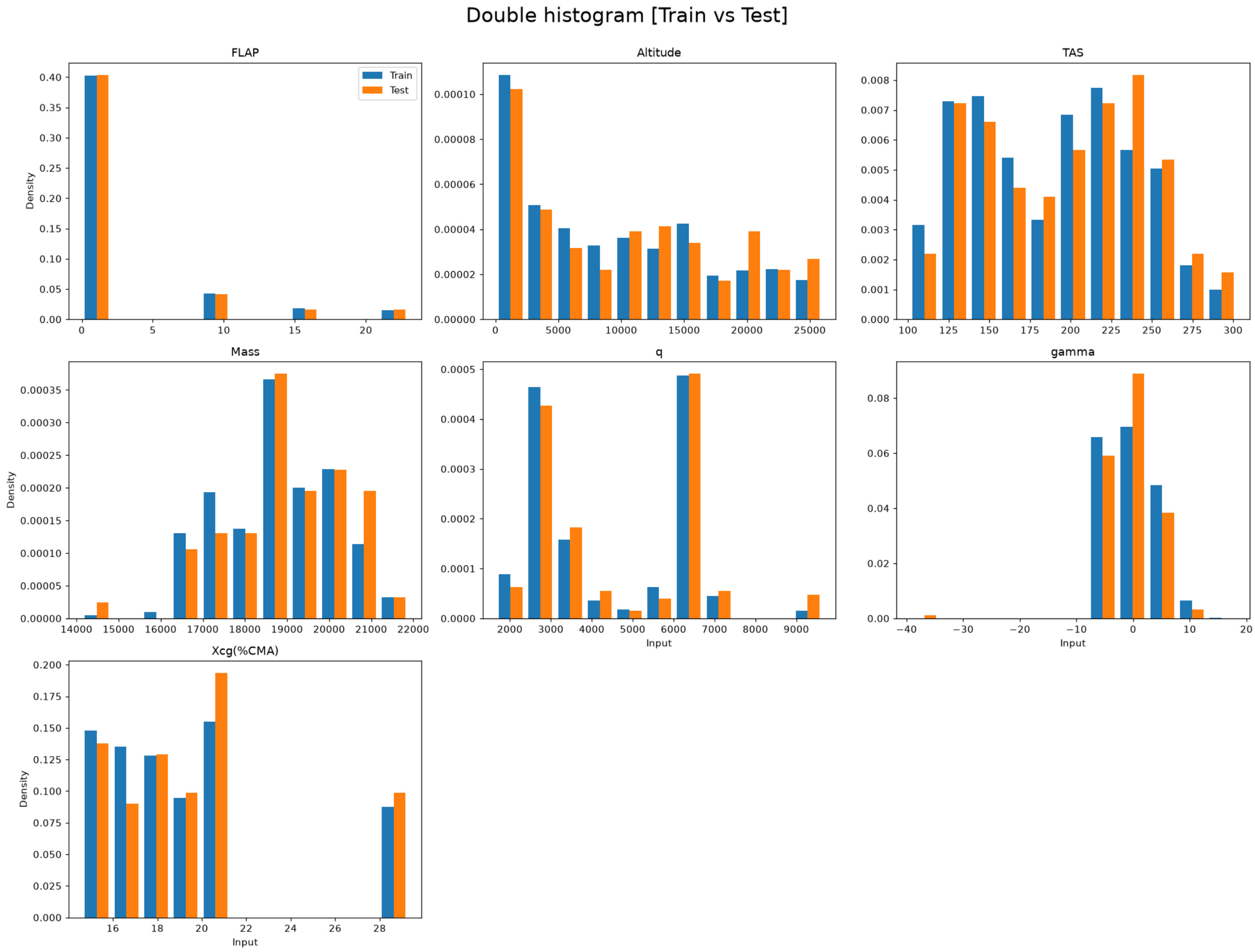
Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0: both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test



KS and AD Test Results

	KS	AD
1g	0.12601607692346975	0.16557574708490383
Vertical maneuver	0.9577252265855336	0.25
Vertical gust	0.4767770596888548	0.25
Turn	0.4440799002733104	0.15119819759103426
Frontal gust	0.9999764926830589	0.25
n0	0.7097390087434631	0.25
Giro	0.4078625477599695	0.25

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
1g	174	-9.66	1.97	-0.210	$^{-0.0561}_{-0.354}$	-0.0456	$^{0.000637}_{-0.127}$	1.16	$^{1.54}_{0.71}$	0.594	$^{0.681}_{0.417}$	29.2	$^{45.7}_{3.37}$	-4.36	$^{-0.403}_{-5.70}$
Vertical maneuver	174	-2.39	1.85	-0.0341	$^{0.0260}_{-0.106}$	-0.00741	$^{0.0192}_{-0.0303}$	0.419	$^{0.535}_{0.326}$	0.224	$^{0.260}_{0.187}$	10.8	$^{16.8}_{6.18}$	-1.61	$^{0.316}_{-2.88}$
Vertical gust	174	-0.127	0.0307	-0.00243	$^{-0.000425}_{-0.00525}$	-0.00131	$^{0.000279}_{-0.00228}$	0.0157	$^{0.0219}_{0.0103}$	0.00965	$^{0.0125}_{0.00848}$	23.7	$^{37.0}_{3.42}$	-3.64	$^{-0.49}_{-4.80}$
Turn	174	-0.840	0.881	-0.0176	$^{0.00651}_{-0.0431}$	-0.00305	$^{0.015}_{-0.0139}$	0.184	$^{0.222}_{0.144}$	0.0983	$^{0.123}_{0.0758}$	8.00	$^{12.9}_{4.45}$	-0.788	$^{1.02}_{-2.19}$
Frontal gust	174	-0.249	0.628	0.00612	$^{0.0171}_{-0.00370}$	0.0000916	$^{0.0000916}_{0.0000916}$	0.068	$^{0.0989}_{0.0330}$	0.0000000000000000139	$^{0.00044}_{0.00}$	48.3	$^{93.0}_{12.9}$	5.57	$^{8.30}_{-1.96}$
n0	174	-0.0583	0.0883	0.00218	$^{0.00541}_{-0.000426}$	0.00135	$^{0.00334}_{-0.00205}$	0.0192	$^{0.0225}_{0.0159}$	0.0150	$^{0.0207}_{0.0133}$	4.62	$^{6.57}_{1.37}$	0.979	$^{1.67}_{0.00112}$
Giro	174	-1.55	1.51	-0.034	$^{0.0144}_{-0.0842}$	-0.0113	$^{0.00616}_{-0.0277}$	0.308	$^{0.387}_{0.236}$	0.142	$^{0.185}_{0.0923}$	11.1	$^{15.0}_{6.61}$	-0.253	$^{2.27}_{-2.45}$

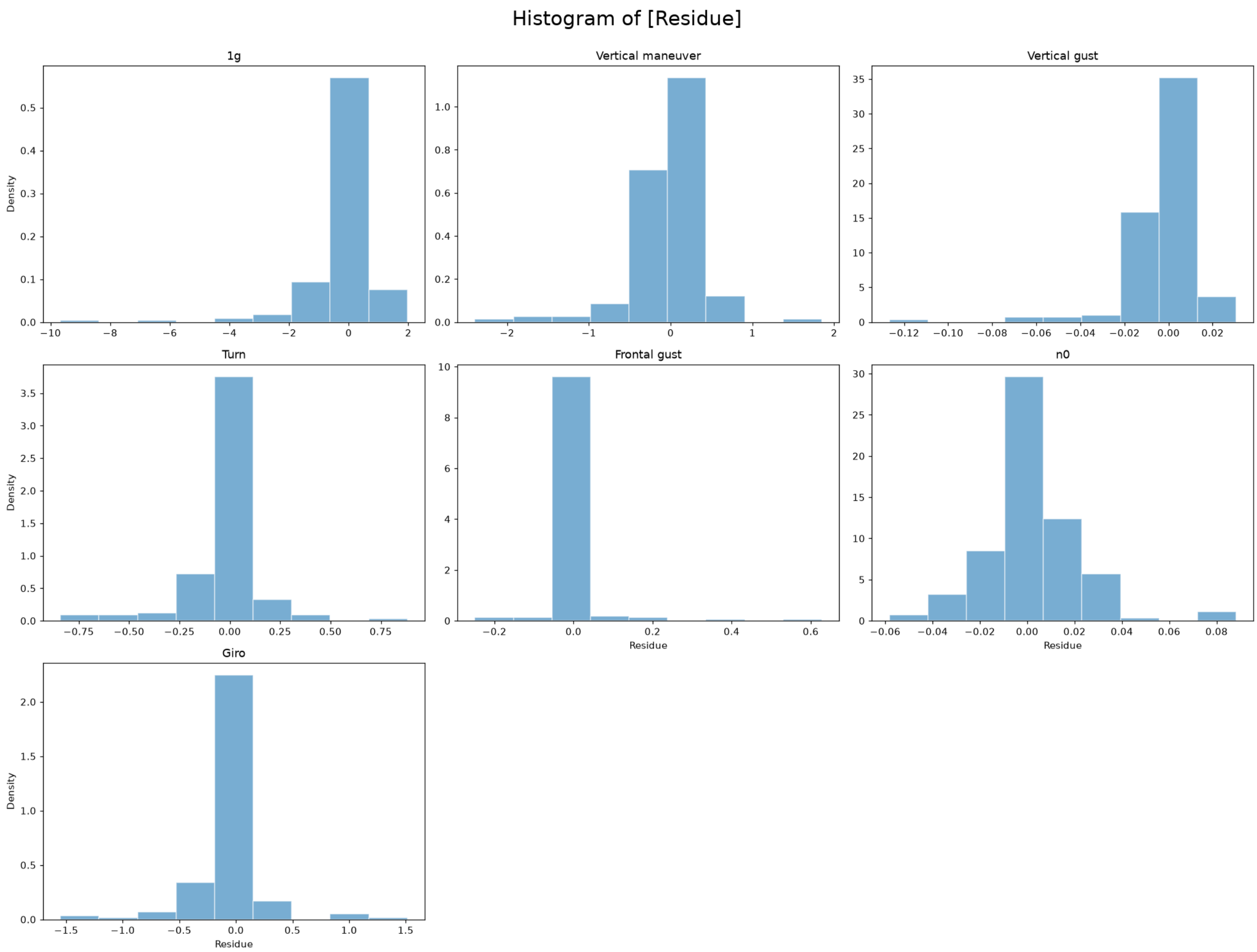
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

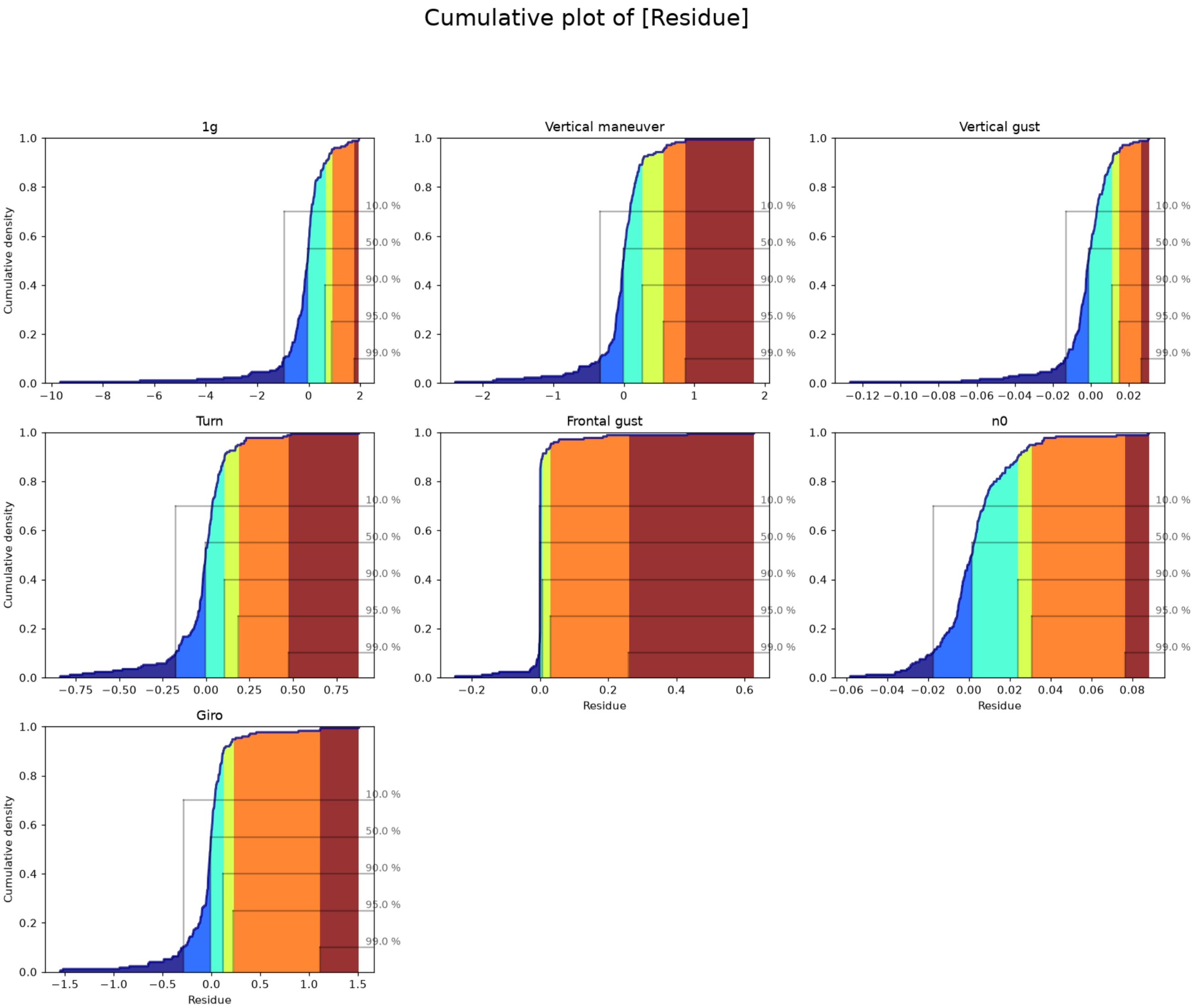
	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
1g	174	0.00282	9.66	0.572	$^{0.701}_{0.448}$	0.246	$^{0.326}_{0.231}$	1.03	$^{1.36}_{0.593}$	0.55	$^{0.655}_{0.418}$	39.7	$^{67.8}_{6.52}$	5.57	$^{7.05}_{2.37}$
Vertical maneuver	174	0.000887	2.39	0.229	$^{0.282}_{0.18}$	0.113	$^{0.132}_{0.0965}$	0.353	$^{0.452}_{0.248}$	0.177	$^{0.260}_{0.142}$	13.9	$^{21.2}_{6.16}$	3.45	$^{4.19}_{2.45}$
Vertical gust	174	0.0000312	0.127	0.00882	$^{0.0115}_{0.00730}$	0.00550	$^{0.00638}_{0.00459}$	0.0132	$^{0.0202}_{0.00751}$	0.00731	$^{0.00935}_{0.00646}$	38.1	$^{60.3}_{6.22}$	5.28	$^{6.60}_{2.12}$
Turn	174	0.00162	0.881	0.105	$^{0.124}_{0.0843}$	0.0513	$^{0.0605}_{0.0366}$	0.153	$^{0.19}_{0.111}$	0.0955	$^{0.133}_{0.0705}$	9.81	$^{14.8}_{5.22}$	3.00	$^{3.57}_{2.36}$
Frontal gust	174	0.0000916	0.628	0.0162	$^{0.0288}_{0.00768}$	0.0000916	$^{0.0000916}_{0.0000916}$	0.0663	$^{0.103}_{0.0283}$	0.00198	$^{0.0057}_{0.000256}$	49.2	$^{90.7}_{14.2}$	6.54	$^{8.82}_{3.82}$
n0	174	0.0000560	0.0883	0.0130	$^{0.0151}_{0.0113}$	0.00763	$^{0.00954}_{0.00593}$	0.0143	$^{0.0175}_{0.0105}$	0.0141	$^{0.0181}_{0.0109}$	8.74	$^{12.7}_{1.10}$	2.52	$^{3.02}_{1.25}$
Giro	174	0.00118	1.55	0.158	$^{0.191}_{0.127}$	0.0569	$^{0.0887}_{0.046}$	0.266	$^{0.326}_{0.196}$	0.149	$^{0.198}_{0.0956}$	12.4	$^{21.5}_{7.42}$	3.36	$^{4.09}_{2.74}$

Absolute-error statistics with bootstrap confidence intervals.

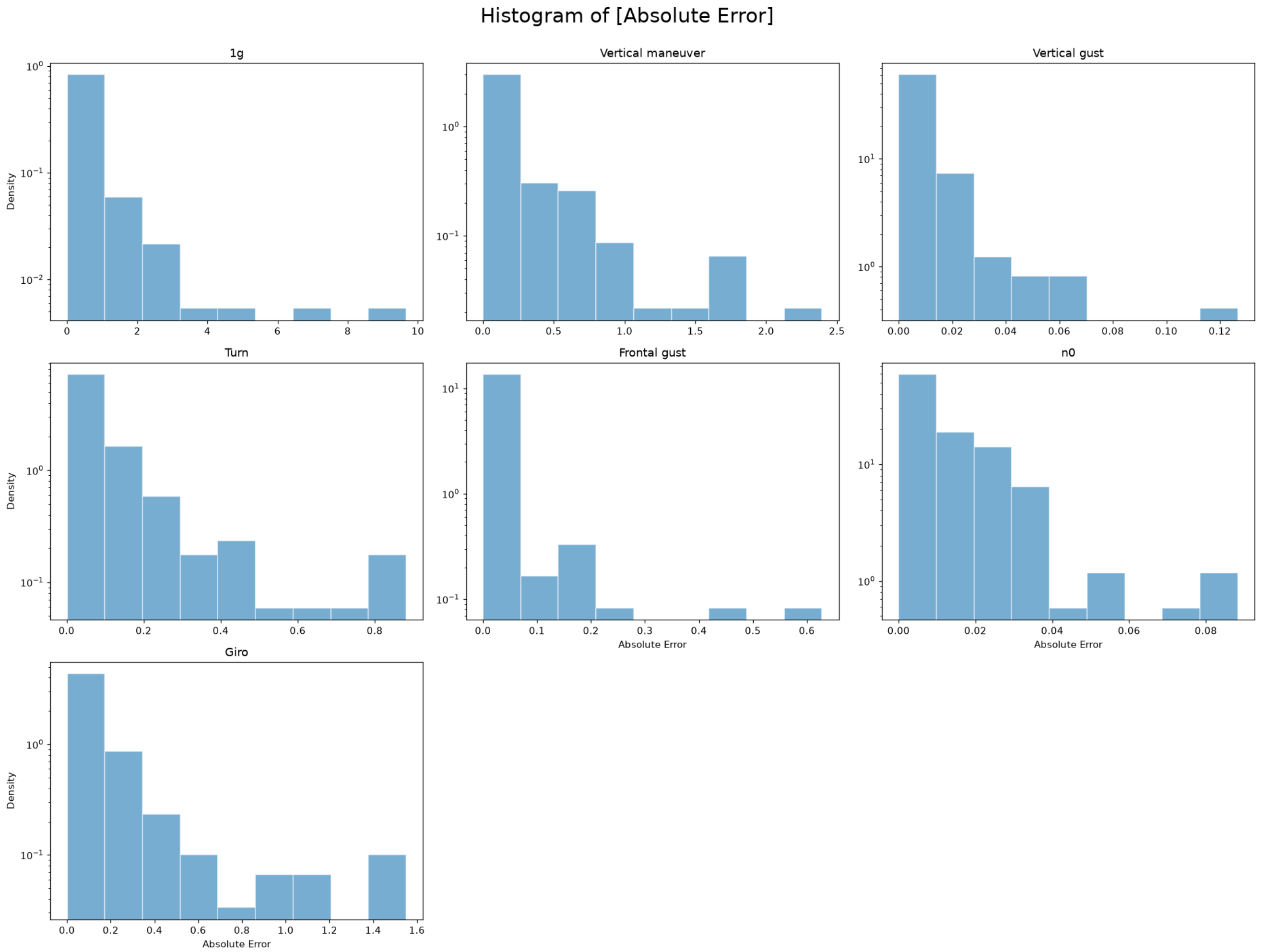
Residue Histograms



Residue CDF

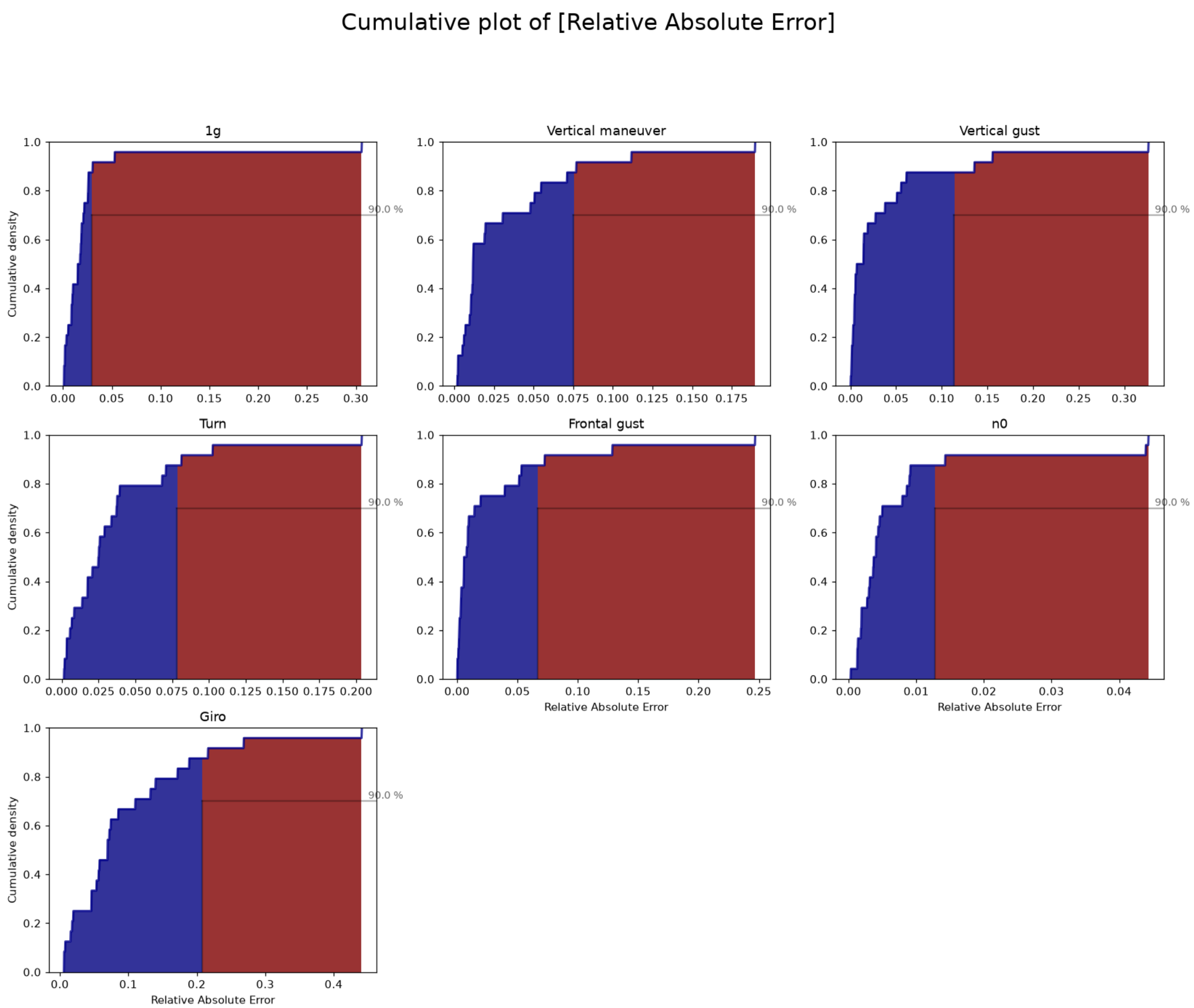


Absolute Error Histograms



Relative Absolute Error CDF — Q90 Accuracy Requirement

Orange dotted = observed Q90; red dashed = target threshold. Green title = PASS; red title = FAIL.

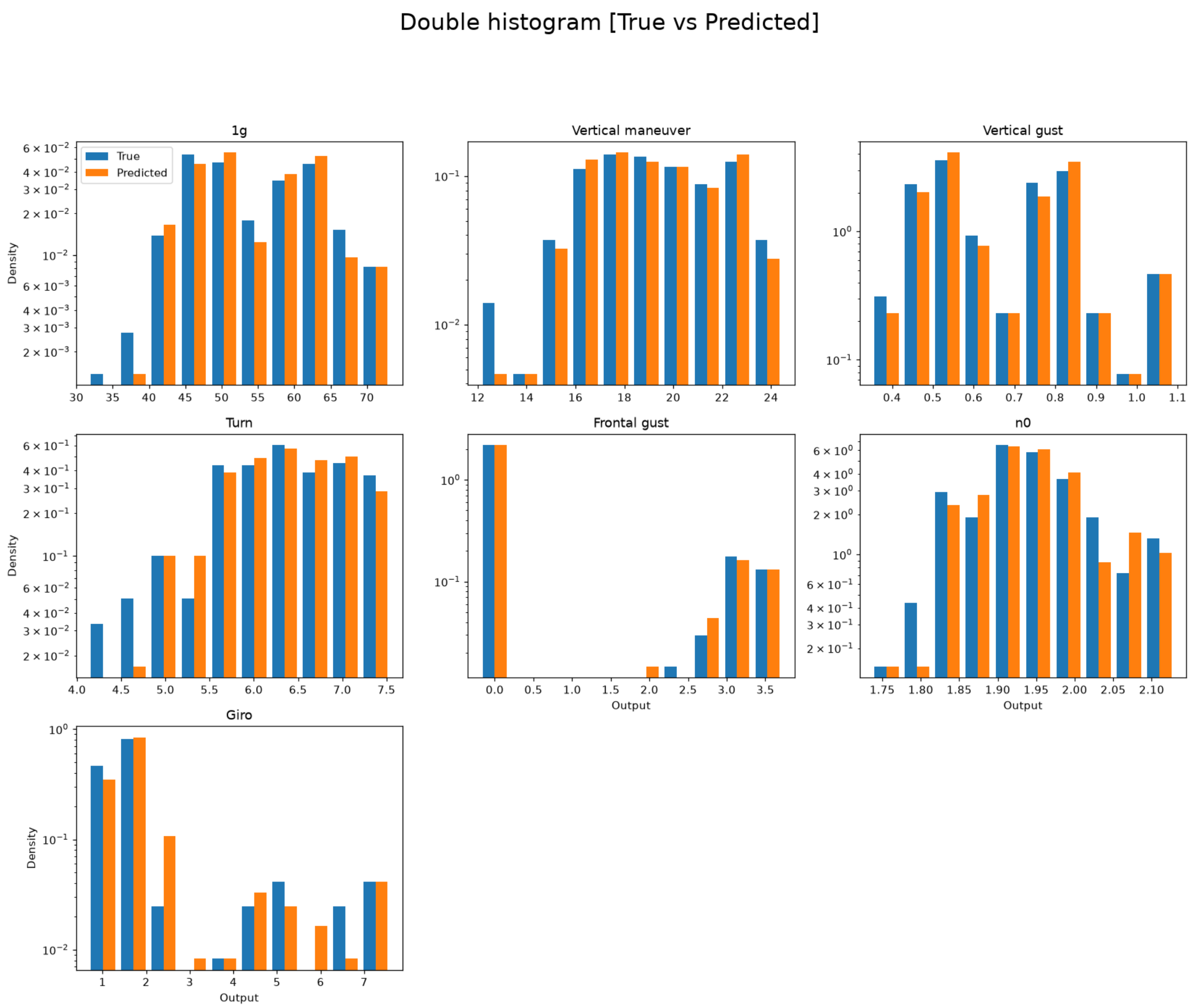


Output	Q90	Target	Status
1g	0.0257	10%	PASS
Vertical maneuver	0.0318	10%	PASS
Vertical gust	0.0267	10%	PASS
Turn	0.0351	10%	PASS
Frontal gust	0.0668	10%	PASS
n0	0.0150	10%	PASS
Giro	0.2236	10%	FAIL

$p \geq 0.05$ — pass $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

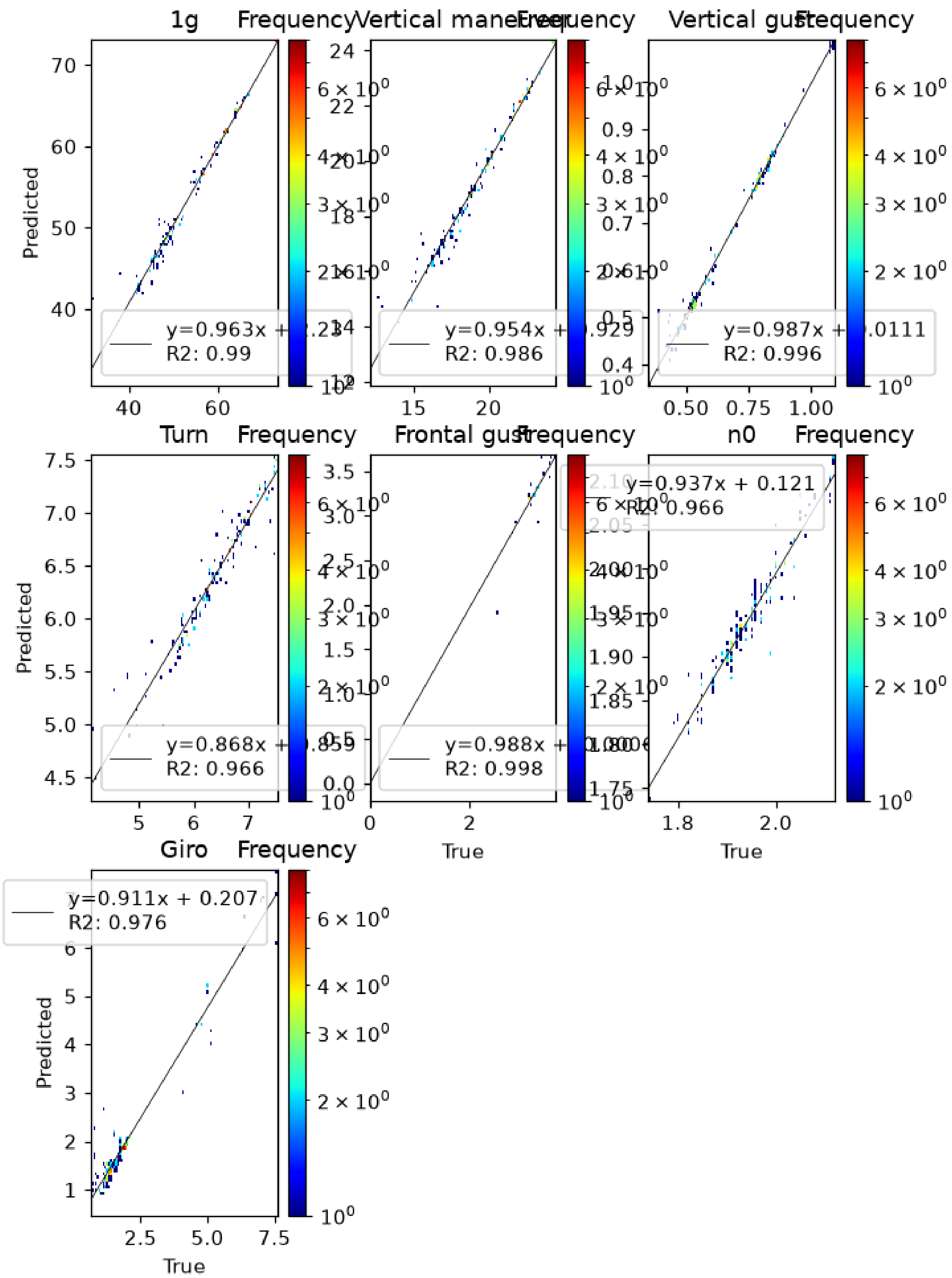


	AD	KS
1g	0.25	0.9931635214778011
Vertical maneuver	0.25	0.9372048405515155
Vertical gust	0.25	0.9747502686921694
Turn	0.25	0.9372048405515155
Frontal gust	0.001	5.39371043839088e-46
n0	0.25	0.6278128269392703
Giro	0.21426024782534803	0.12392282795362734

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.


1g
Frequency
Vertical manoeuvre
Frequency
Vertical gust
Frequency
Turn
Frequency
Frontal gust
Frequency
n0
Frequency
Giro
Frequency
Predicted
True
y=0.963x + 1.23
R2: 0.99
y=0.954x + 0.929
R2: 0.986
y=0.987x + 0.0111
R2: 0.996
y=0.868x + 0.859
R2: 0.966
y=0.988x + 0.0006
R2: 0.998
y=0.937x + 0.121
R2: 0.966
y=0.911x + 0.207
R2: 0.976

P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: FLAP→Turn, Mass→1g, Mass→Frontal gust, Mass→n0, q→1g, q→Vertical maneuver, q→Vertical gust, q→Turn, q→Giro, gamma→Frontal gust…. These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

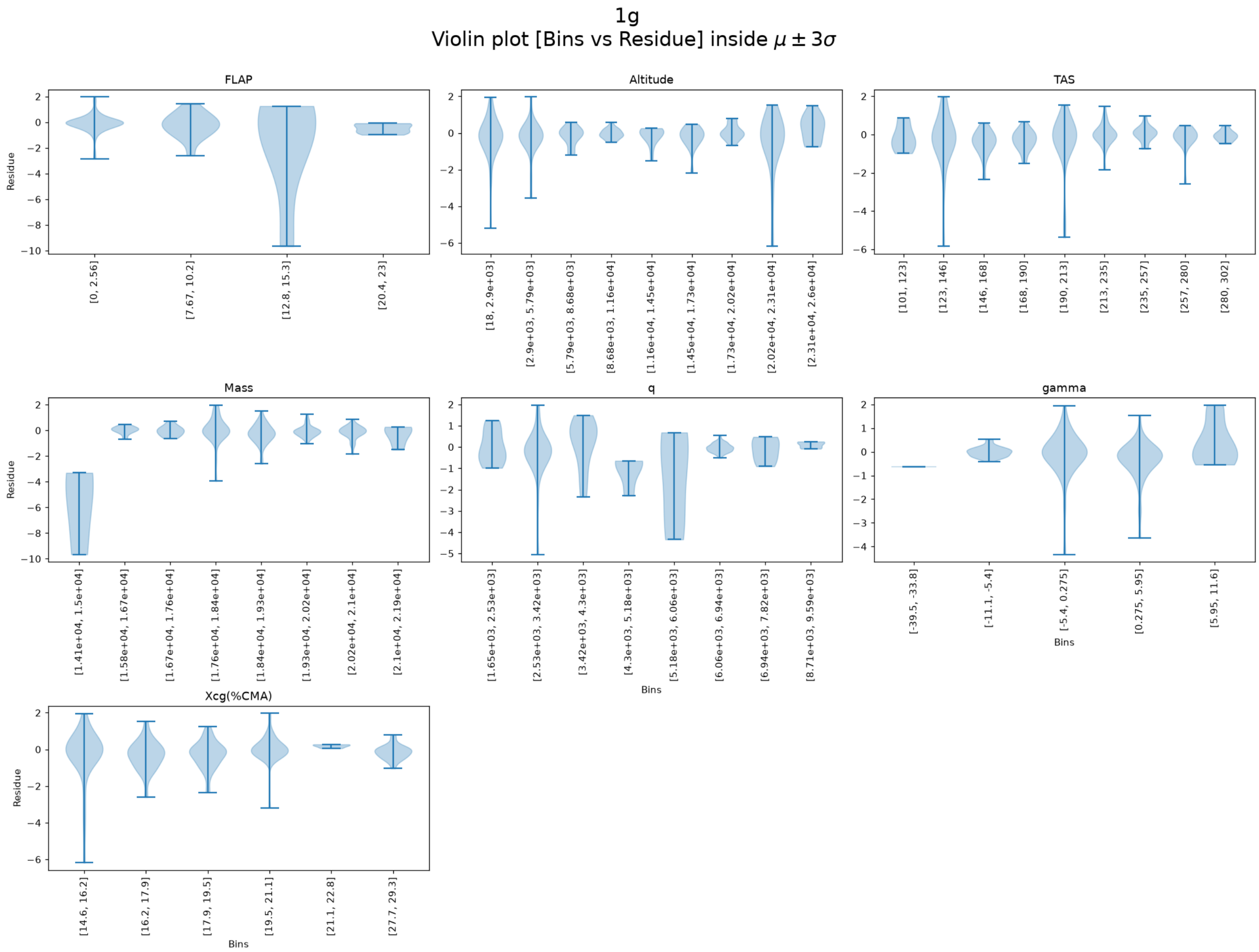
	1g	Vertical maneuver	Vertical gust	Turn	Fronta
FLAP_binned	0.2065973724768399	0.06104011718038691	0.15668564248348774	0.005793851223069571	0.10148153552
Altitude_binned	0.7600647376528422	0.6685726985430076	0.20770302663221002	0.4911758164166914	0.97938992331
TAS_binned	0.373883876500768	0.8346350123713844	0.7358769291681275	0.07180184139864736	0.5515465774
Mass_binned	0.03979950129289494	0.13958767162222122	0.7809502379154429	0.08975212620353369	0.00100533963469
q_binned	0.029142840681280292	0.004109515764661657	0.004352593886667565	0.04687166937364112	0.118139389541
gamma_binned	0.0573856197493953	0.6324324031842394	0.7522158259107165	0.25033324446048494	0.00228091600036
Xcg(%CMA)_binned	0.047851843523705626	0.4163741104766572	0.39974574684579645	0.3511065757629104	0.0087199032100

Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

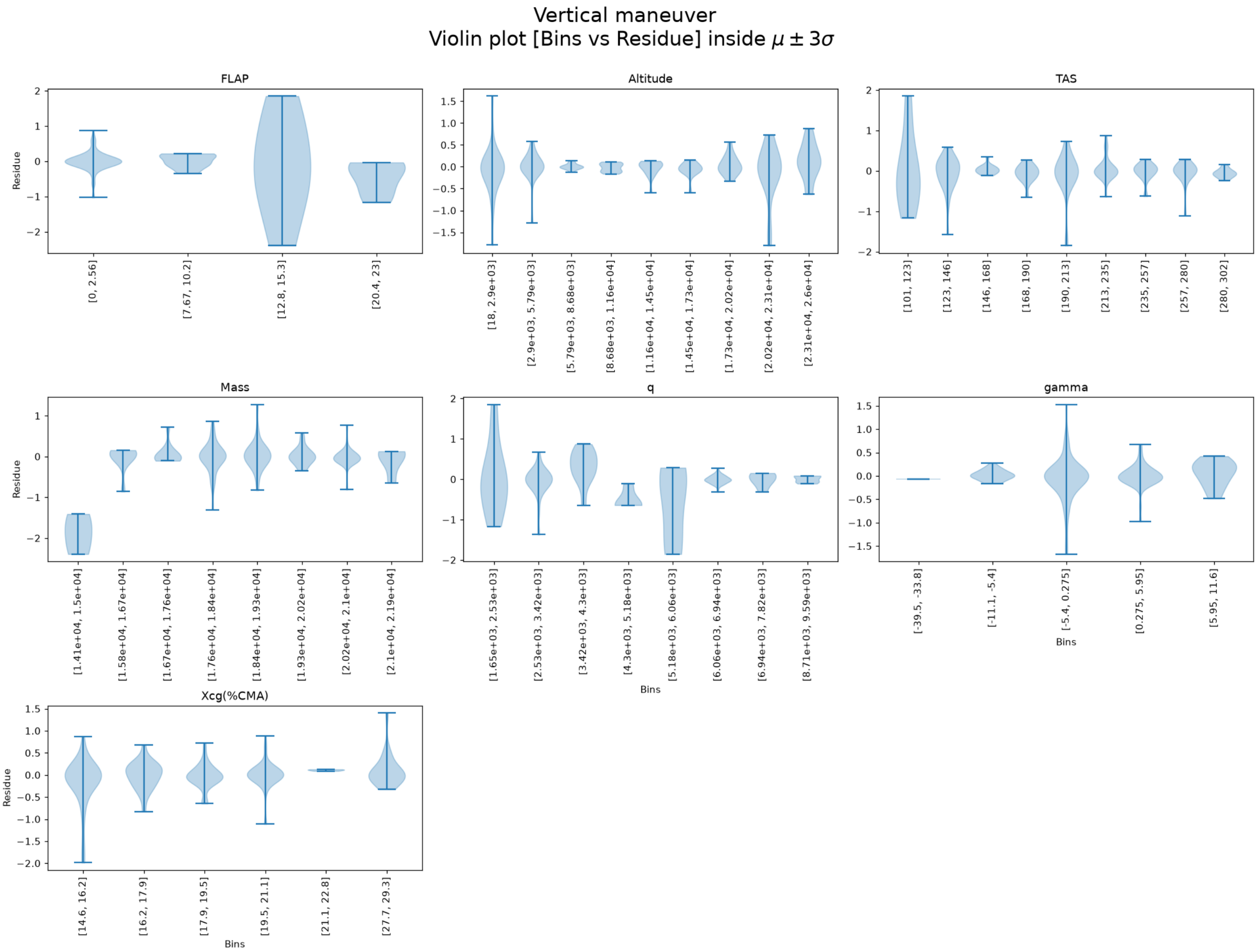
Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.

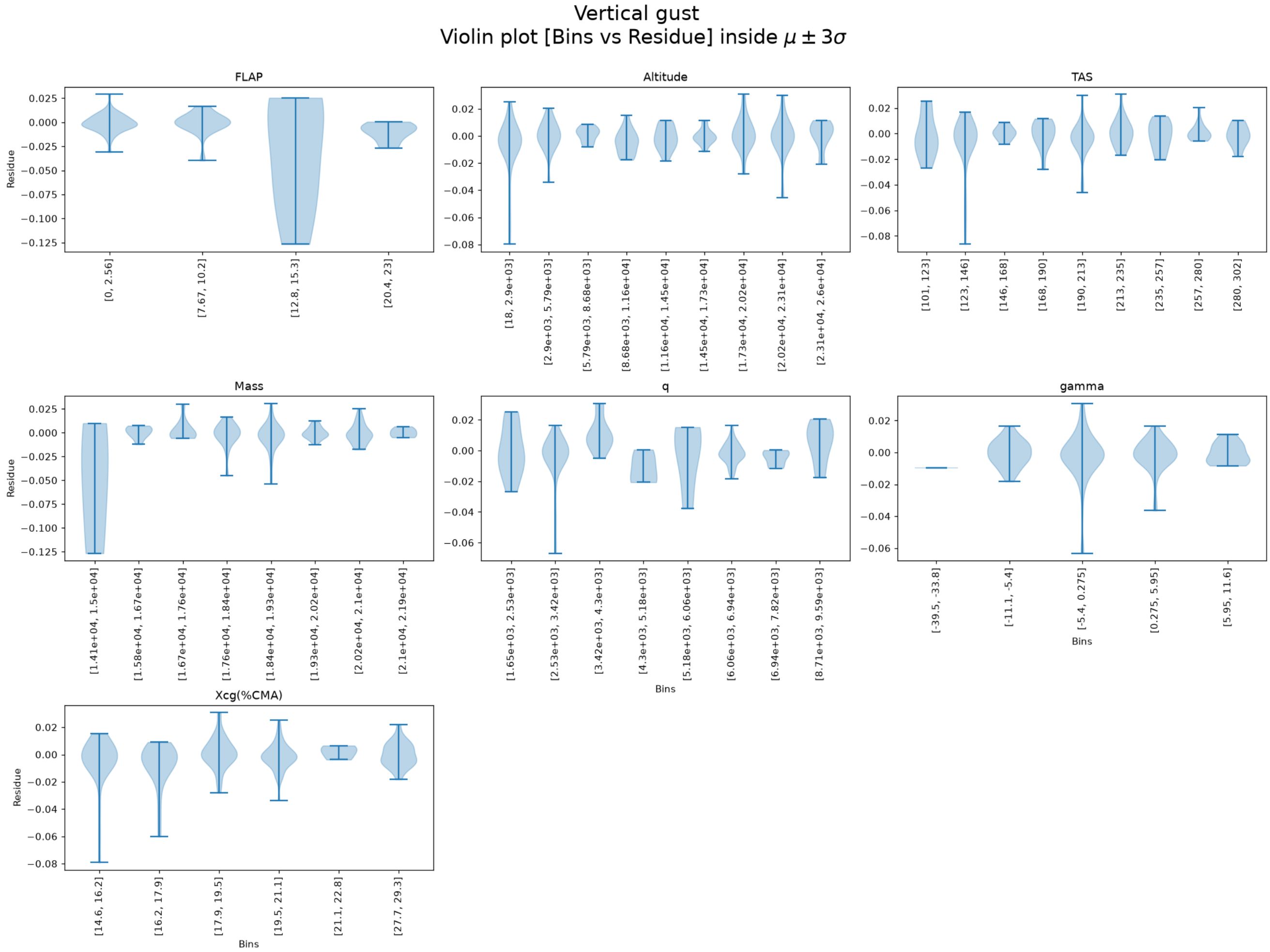
Residue — 1g



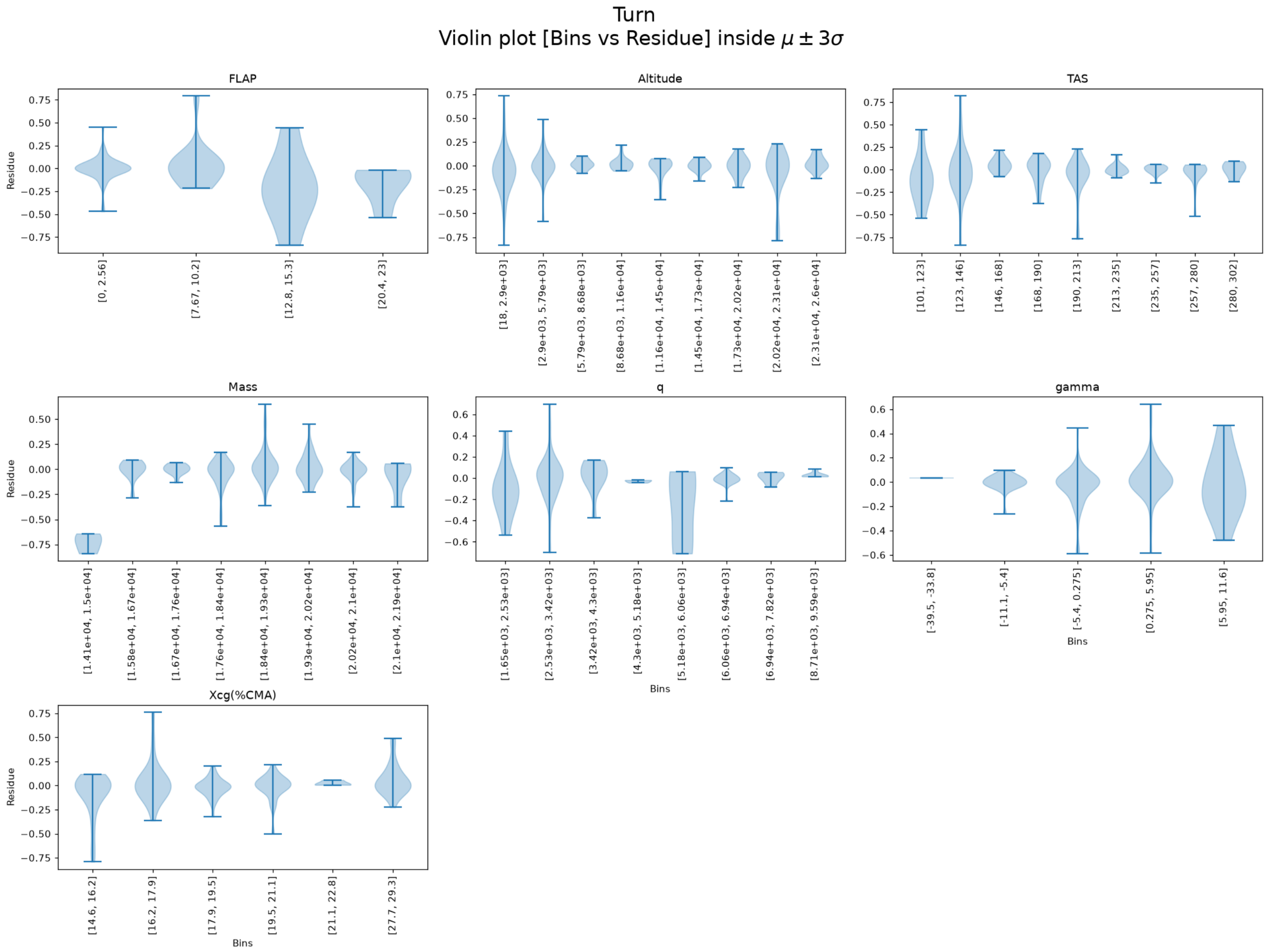
Residue — Vertical maneuver



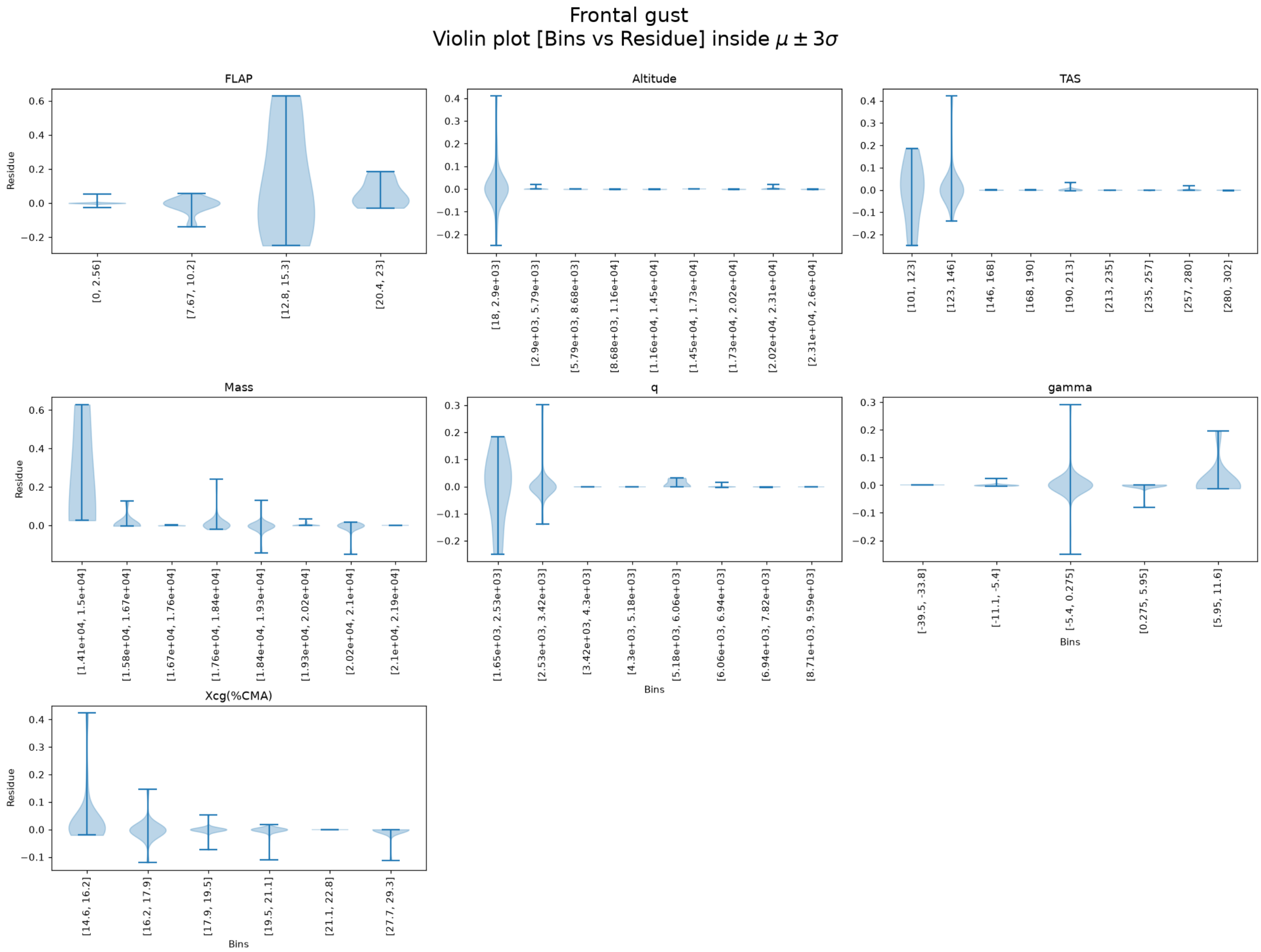
Residue — Vertical gust



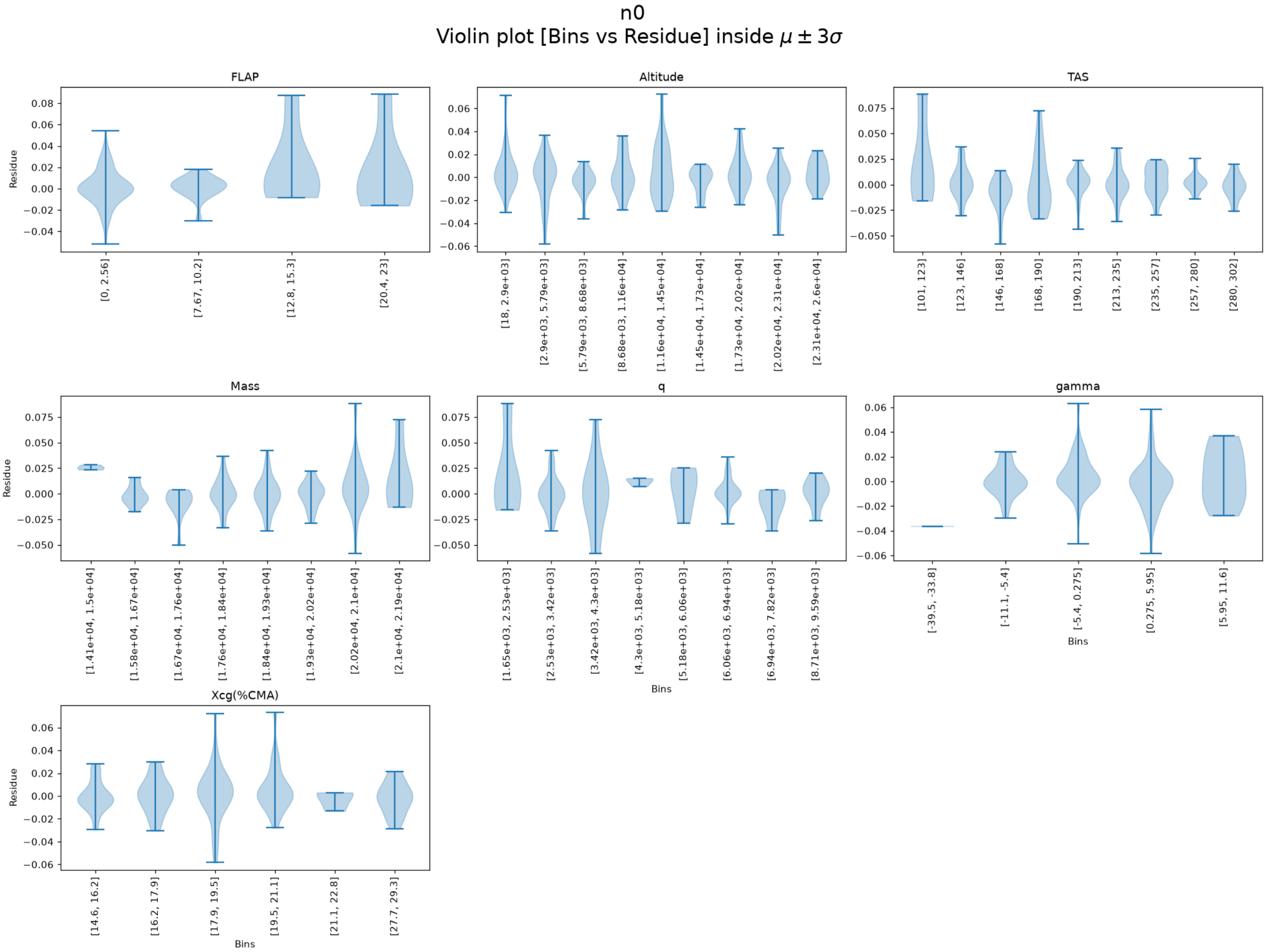
Residue — Turn



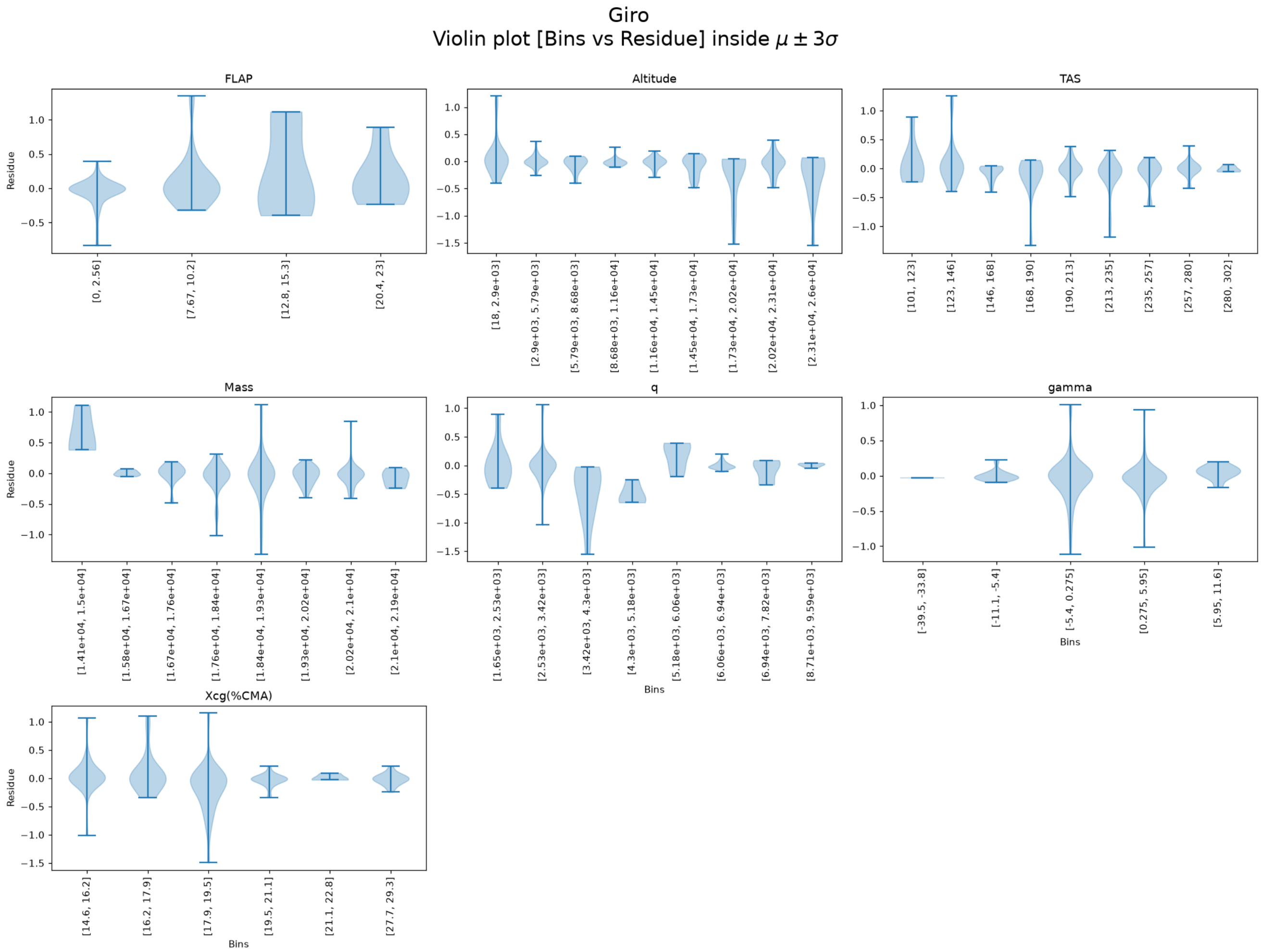
Residue — Frontal gust



Residue — n0

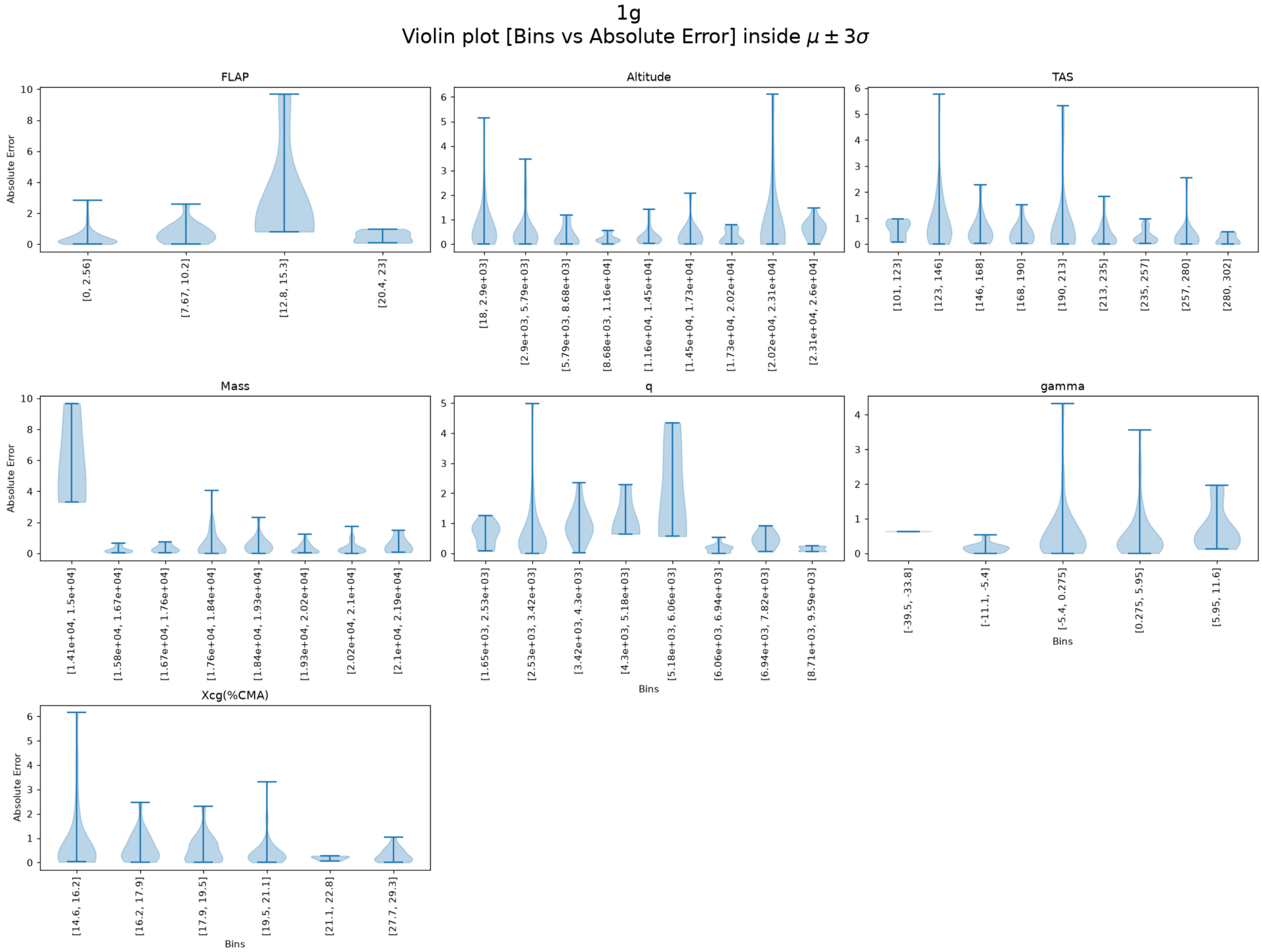

n0
Violin plot [Bins vs Residue] inside $\mu \pm 3\sigma$
FLAP
Altitude
TAS
Mass
q
gamma
Xcg(%CMA)
Residue
Bins

Residue — Giro

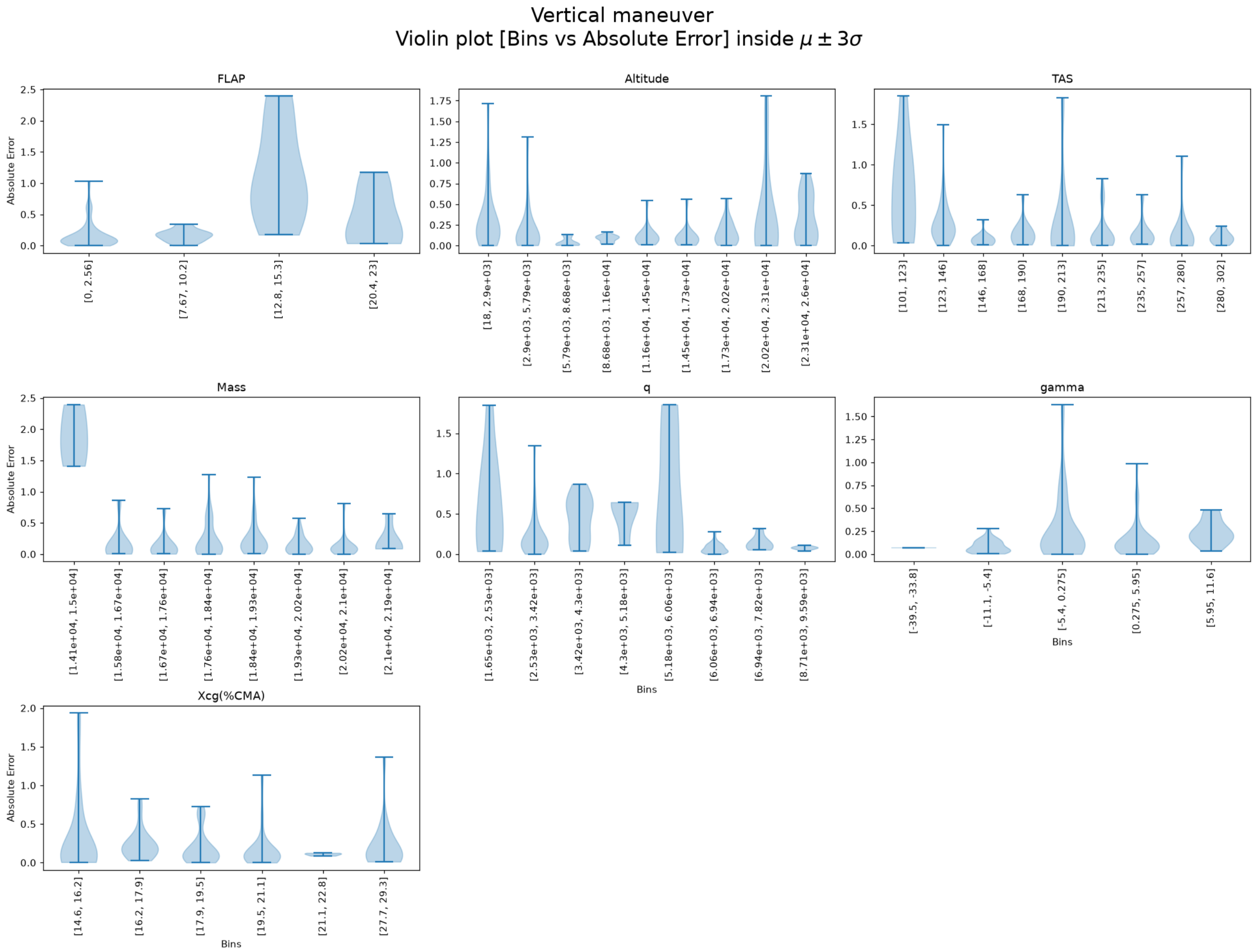


Absolute Error Violin Plots per Output (binned by input)

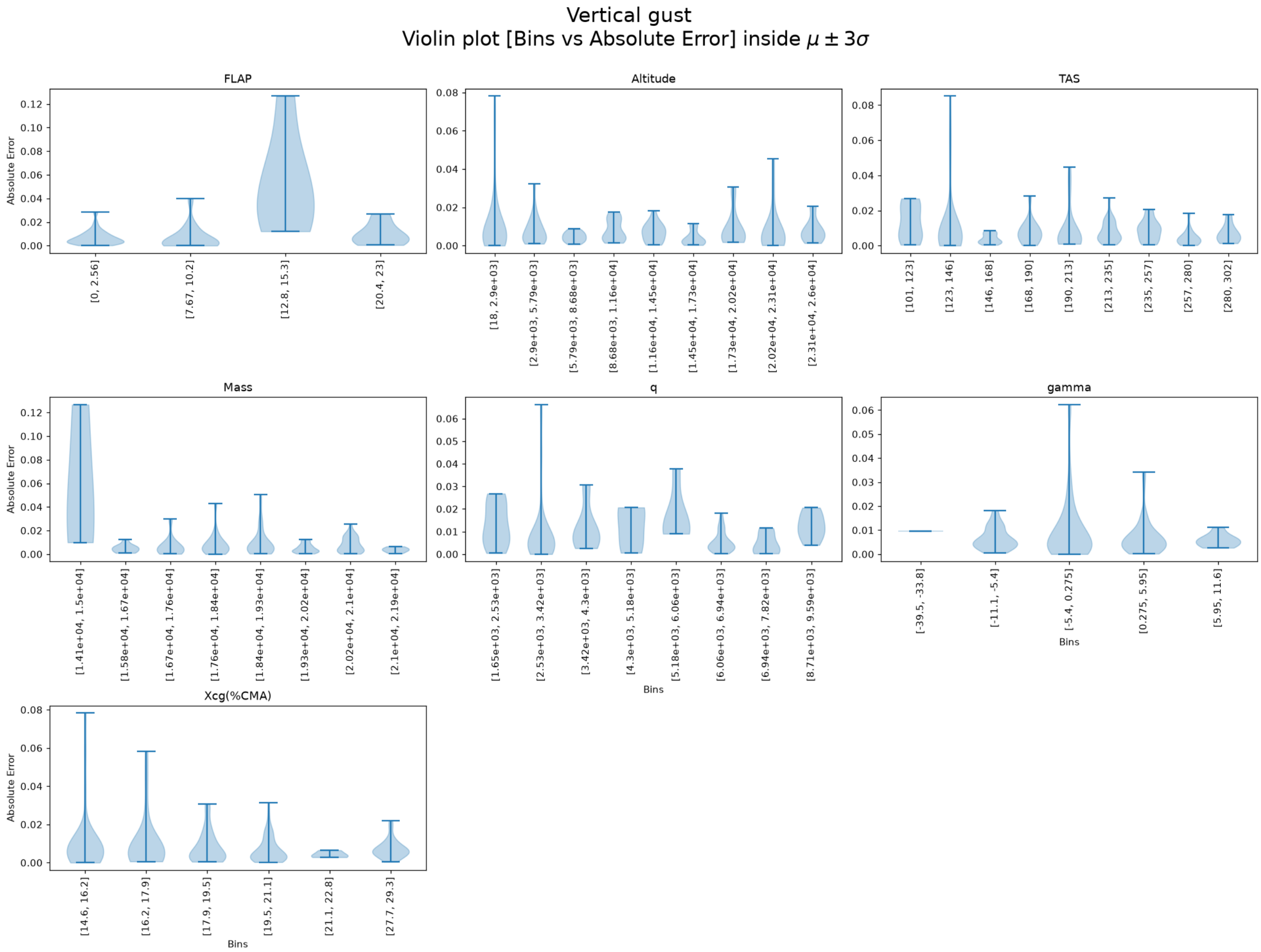
Absolute Error — 1g



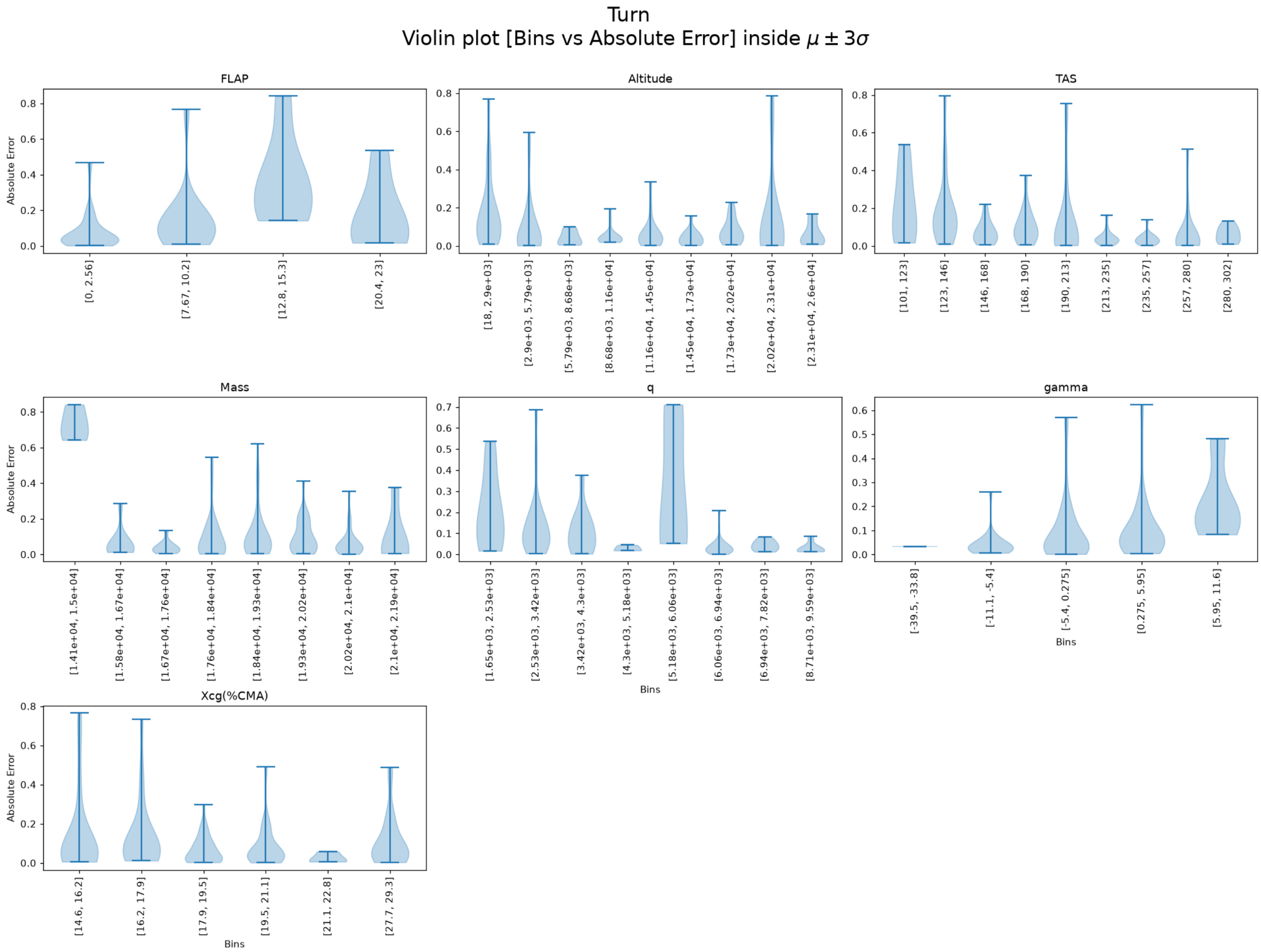
Absolute Error — Vertical maneuver



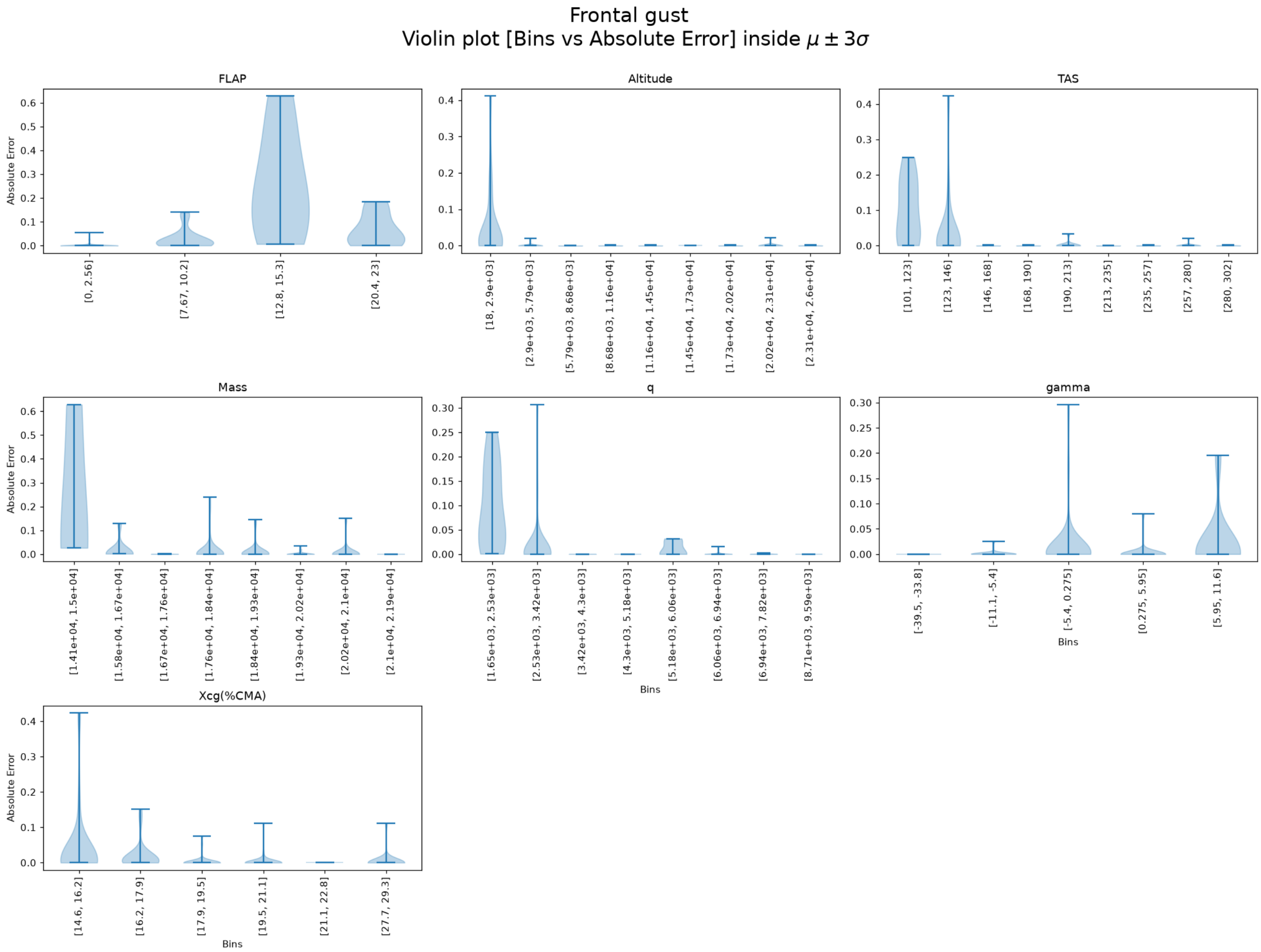
Absolute Error — Vertical gust



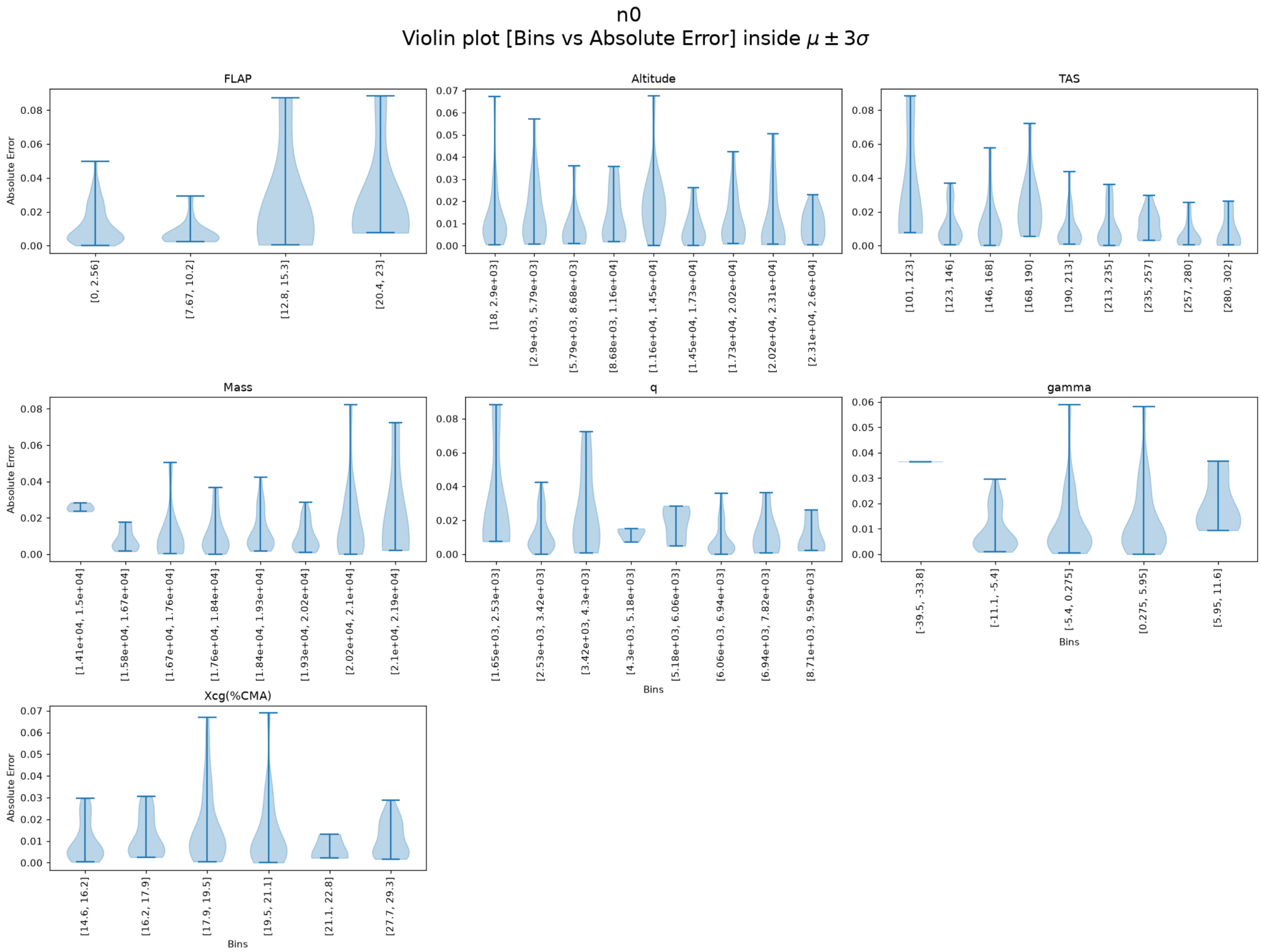
Absolute Error — Turn



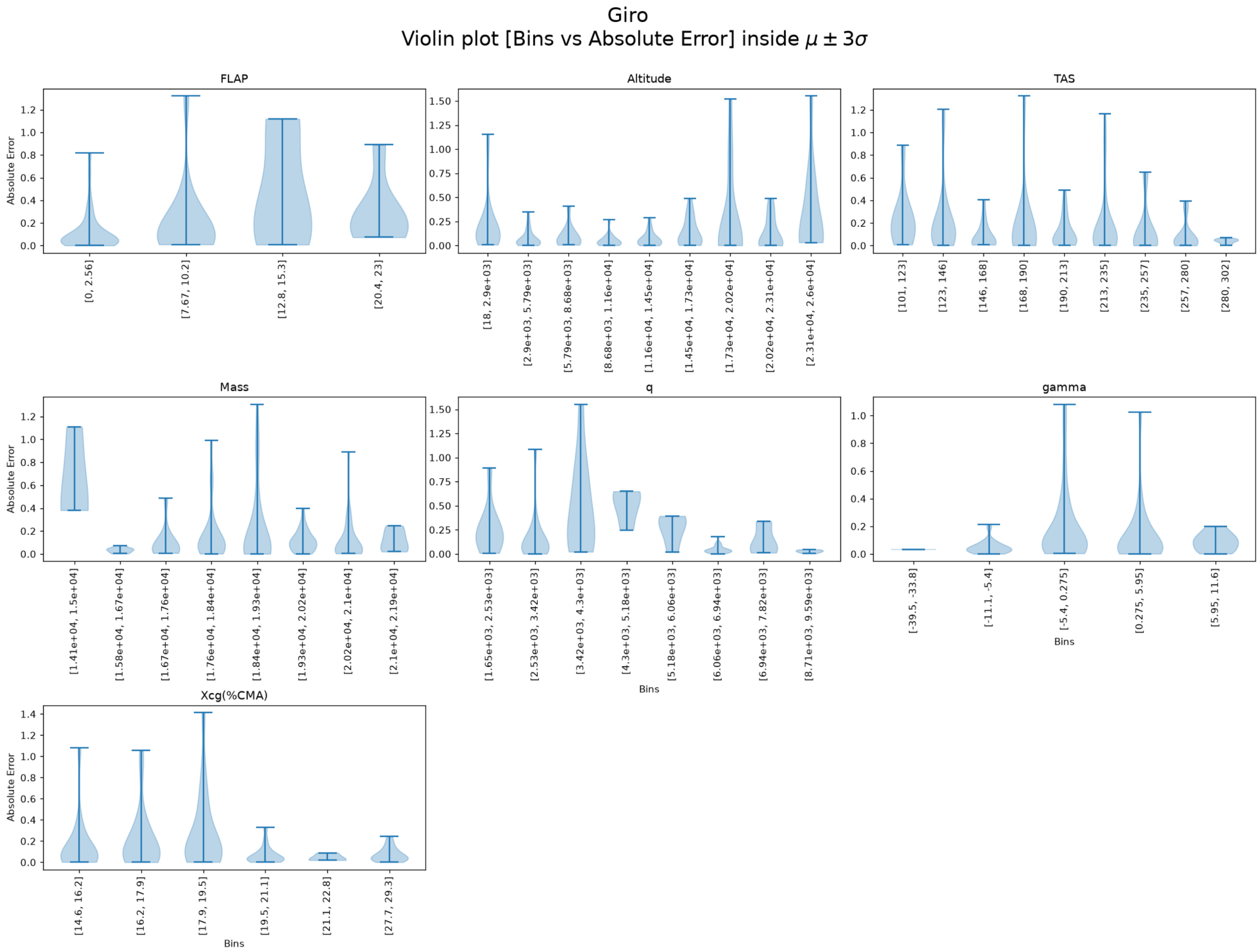
Absolute Error — Frontal gust



Absolute Error — n0



Absolute Error — Giro

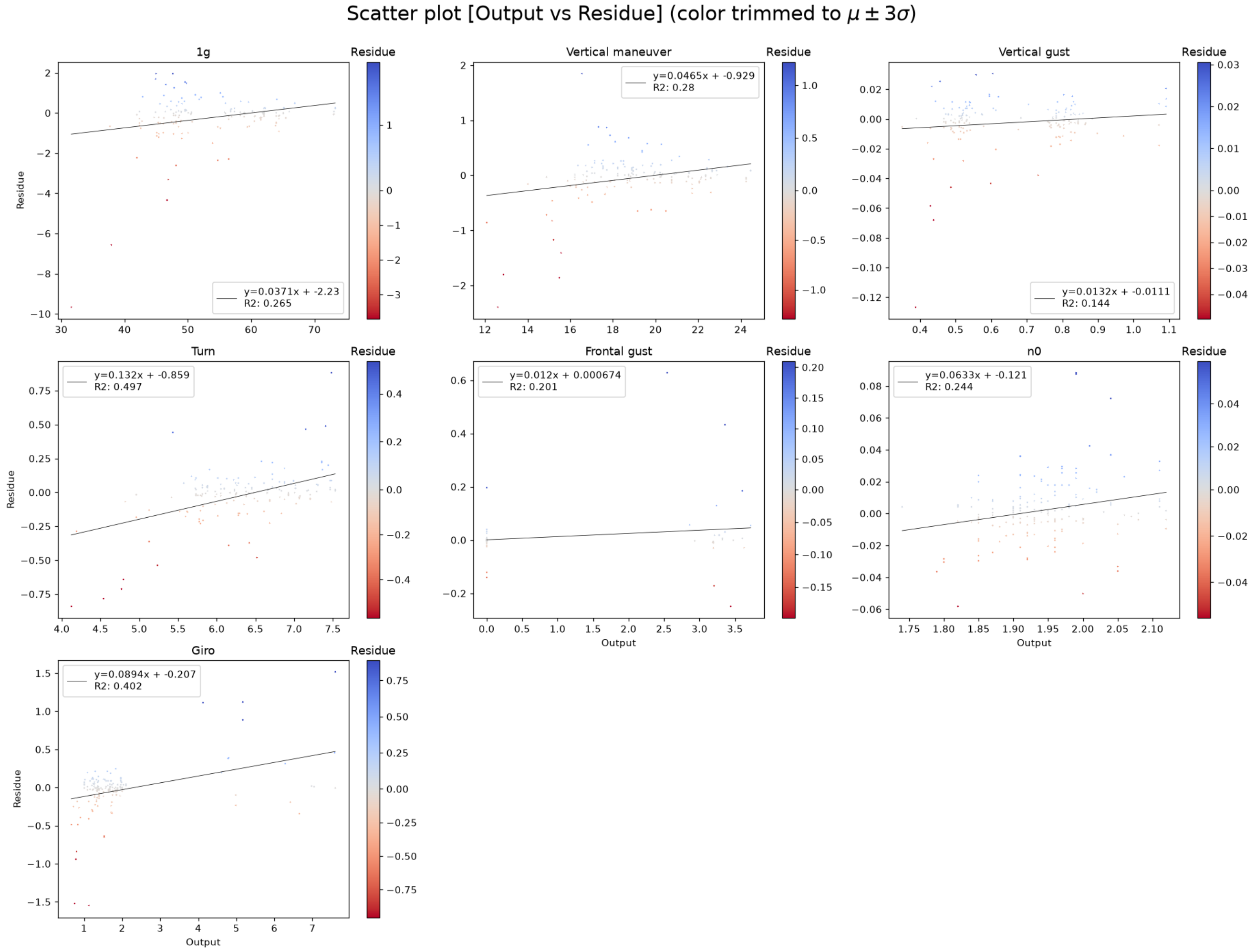


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

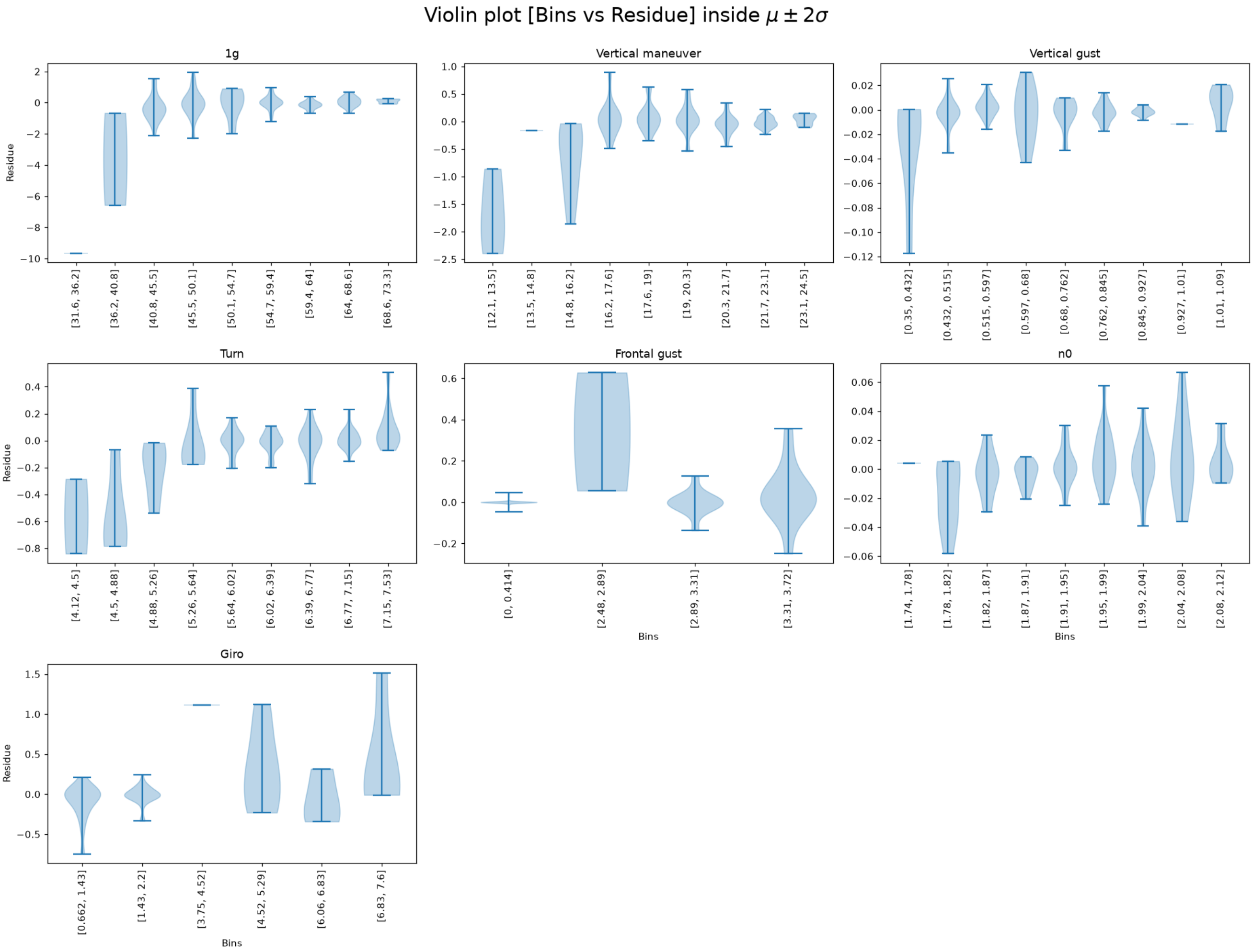
Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.



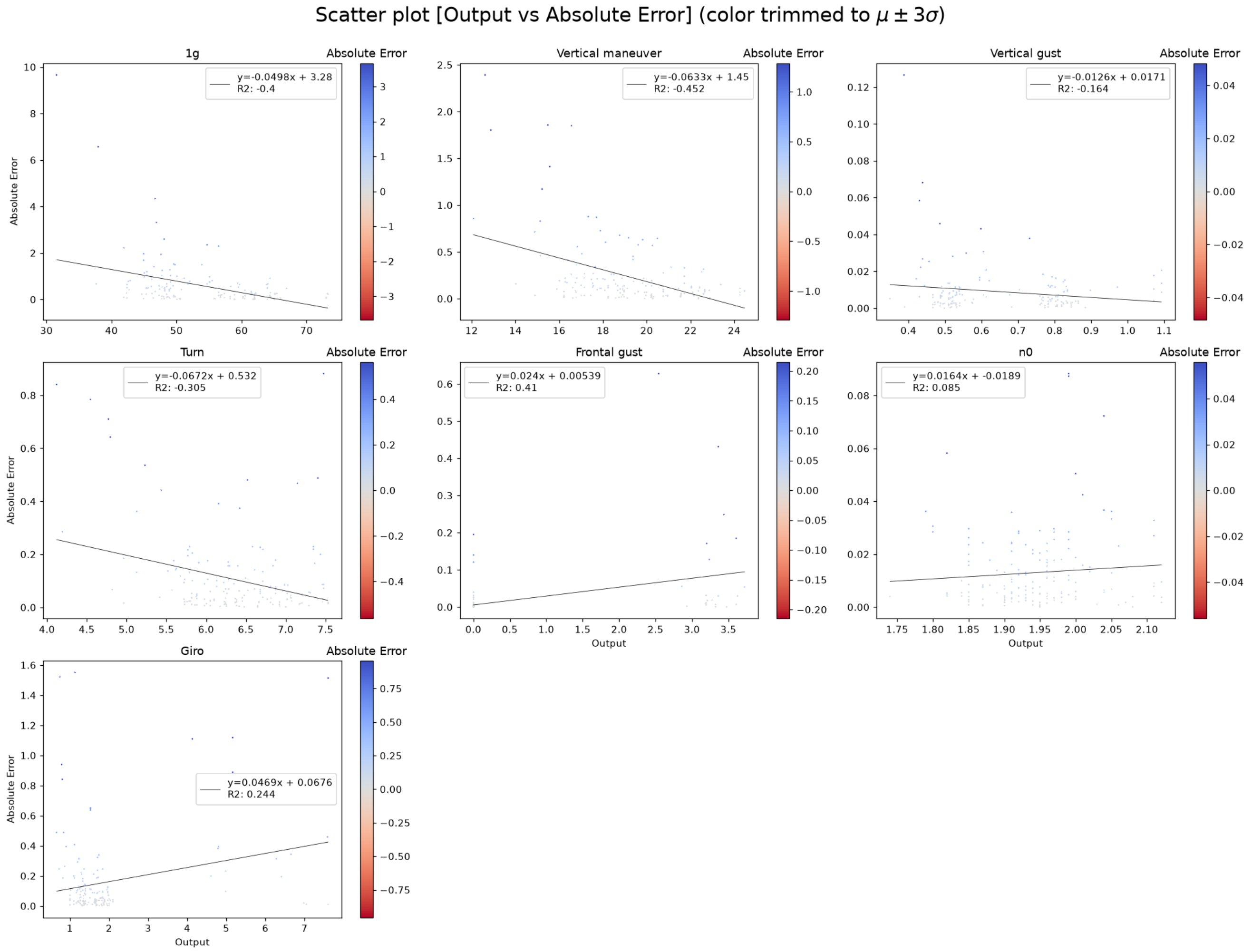
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

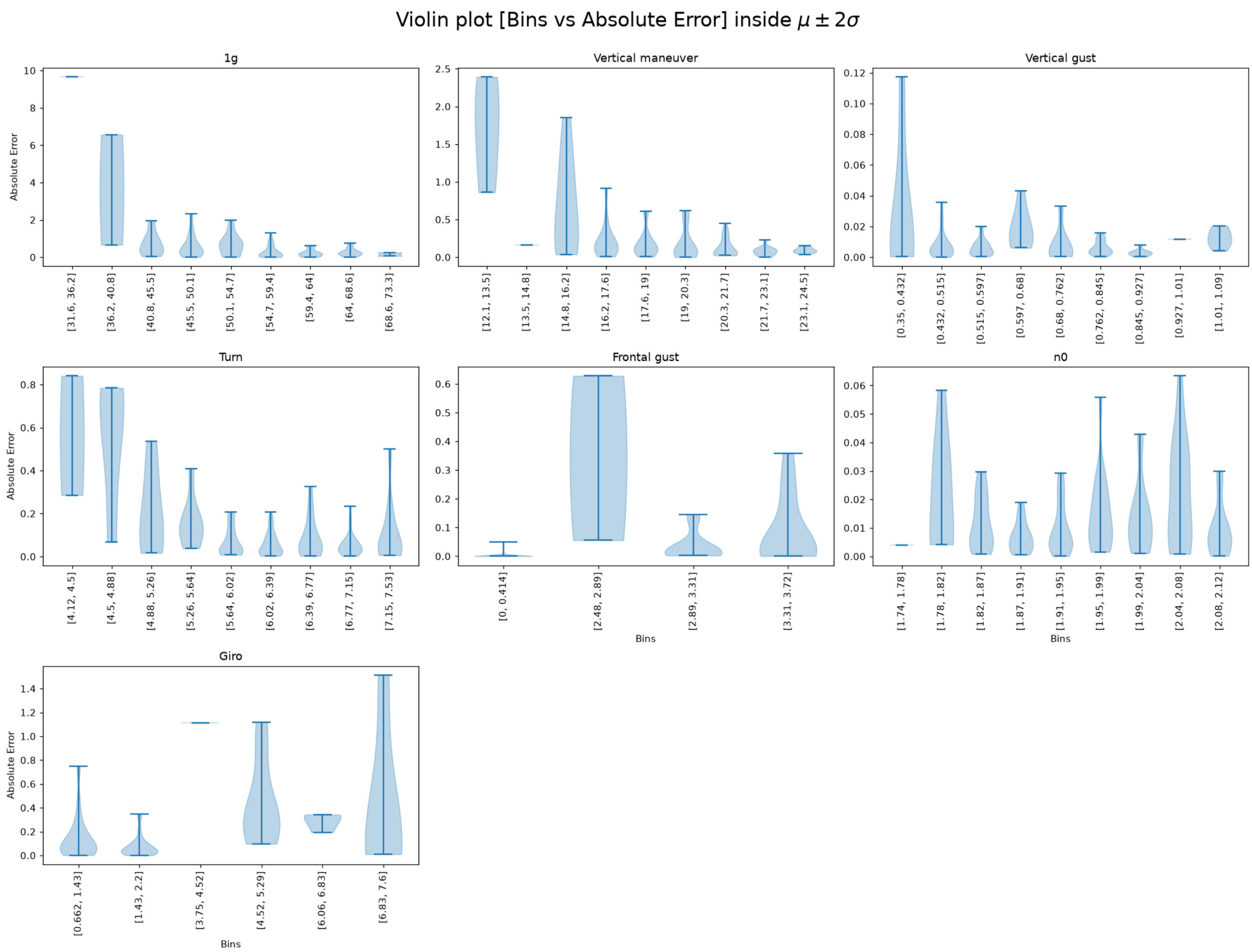


Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.



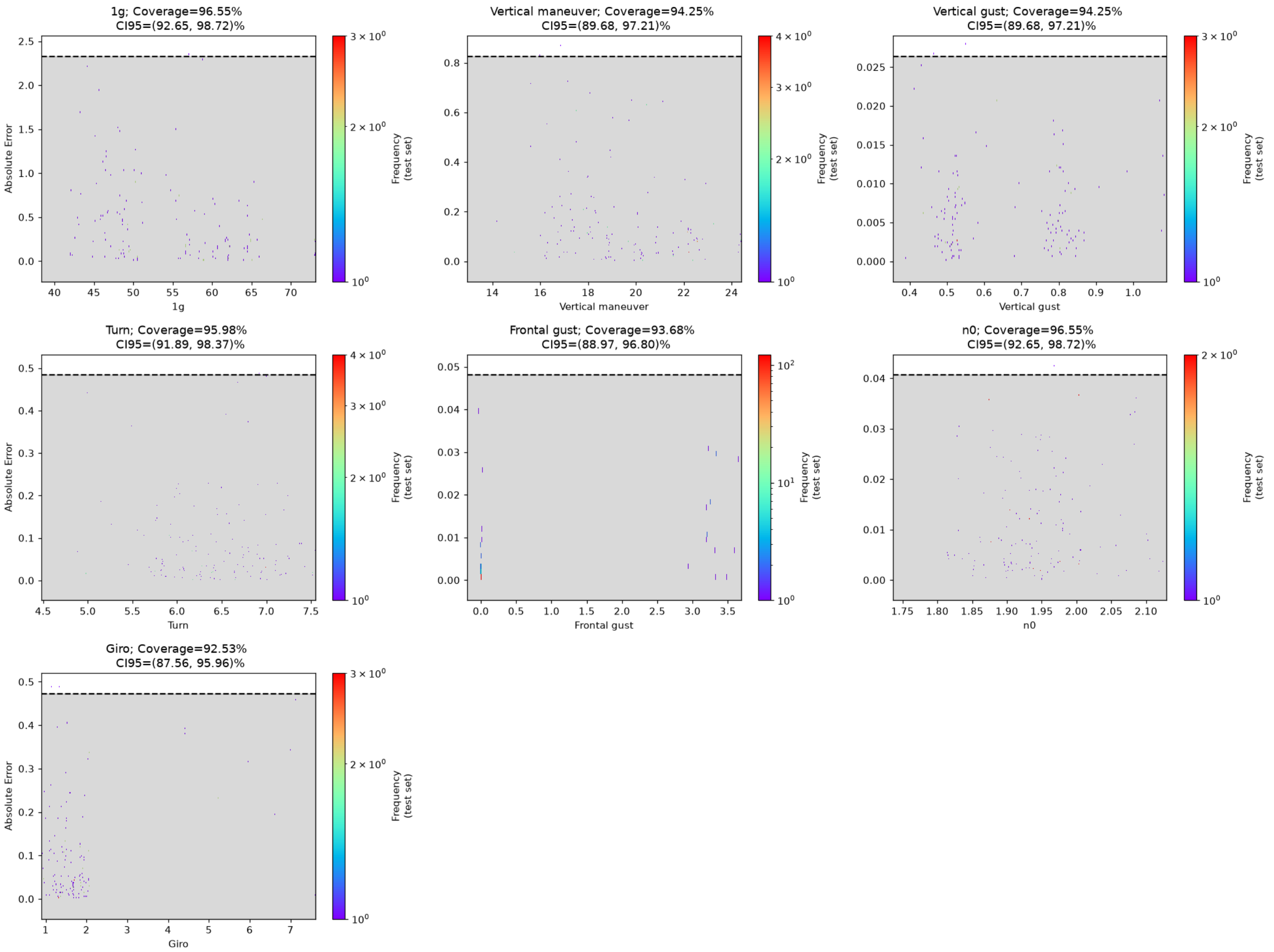
Absolute Error Violin vs True Output Bins


Violin plot [Bins vs Absolute Error] inside $\mu \pm 2\sigma$
1g
Vertical maneuver
Vertical gust
Turn
Frontal gust
n0
Giro
Absolute Error
Bins

Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	1g	Vertical maneuver	Vertical gust	Turn	Frontal gust	n0	Giro	TOTAL COV
Coverage	96.55	94.25	94.25	95.98	93.68	96.55	92.53	nan
CI	[92.65, 98.72]	[89.68, 97.21]	[89.68, 97.21]	[91.89, 98.37]	[88.97, 96.8]	[92.65, 98.72]	[87.56, 95.96]	nan

Empirical coverage of the 95th-percentile uncertainty bound on the test split.